A FURTHER RESULTS

This appendix collects the results that Secs. 3, 4, 5 and 6 refer to.

Unsound BCNF decompositions. The following proposition formalizes the observation from Ex. 3.5.

Proposition A.1 (Unsound BCNF decompositions). *There is a schema (R, Σ) and a lossless decomposition $\mathcal{D}$ of (R, Σ) into BCNF, computed by the classical decomposition algorithm, such that the key design $(\mathcal{D}, \kappa)$ that declares the minimal keys of $(S, \Sigma[S])$ on every $S \in \mathcal{D}$ is not sound.*

Inheritance. The following lemma shows that an uncritical FD stays uncritical when other FDs are left unenforced, which Thm. 3.22 and Alg. 4 rest on.

Lemma A.2 (Inheritance). *Let $\mathcal{E} \subseteq \Sigma^+$ be a closed FD set that contains every uncritical minimal FD of Σ. Then every uncritical minimal FD of Σ is an uncritical minimal FD of $\mathcal{E}$.*

Verification. The hardness of Thm. 4.5 rests on the representation of a design by its declared keys. A design is locally enforced iff, for every minimal FD $X \to A$ of $\mathcal{E}$ and every $S \in \mathcal{D}$ with $XA \subseteq S$, the set X contains some $K \in \kappa(S)$, which takes time polynomial in the size of the atomic cover of $\mathcal{E}$.

Shapes of weakened FDs. The following proposition shows that the enumeration of line 5 of Alg. 1 can be confined neither to one weakened form per unenforced FD nor to the left-hand sides of minimal FDs.

Proposition A.3 (Shapes of weakened FDs). *There is a schema (R, Σ) with $\lambda(R, \Sigma) = 1$ such that every BCNF design of (R, Σ) with loss one enforces two weakened forms of the same unenforced minimal FD. There is a schema (R, Σ) with $\lambda(R, \Sigma) = 2$ such that every BCNF design of (R, Σ) with loss two enforces a weakened form $X' \to A$ of an unenforced minimal FD where X' is not contained in the left-hand side of any minimal FD of Σ.*

The closures of the left-hand sides of critical minimal FDs do not confine the weakened FDs either. For $\Psi = \Sigma_a - \Delta$, call a weakened FD $X' \to A$ *local* iff $X' \subseteq Y^+_\Psi$ for some minimal FD $Y \to B$ of Ψ^+ that is critical for Ψ^+. Only the critical minimal FDs of Ψ^+ obstruct the enforceability of Ψ^+, and local weakened FDs suffice in the two schemata of Prop. A.3, but the following proposition shows that they do not suffice in general.

Proposition A.4 (Local weakened FDs). *There is a schema (R, Σ) with a set $\Delta \subseteq \Sigma_a$ and weights w such that $|\Delta| = \lambda(R, \Sigma)$, every BCNF design of (R, Σ) with loss $\lambda_w(R, \Sigma)$ leaves exactly Δ unenforced, and every BCNF design of (R, Σ) that leaves exactly Δ unenforced enforces a weakened FD that is not local.*

Hence, line 5 of Alg. 1 cannot be confined to local weakened FDs. The search of Lem. A.14 needs no such restriction, since it measures closures against the current set $\mathcal{E}$ rather than Ψ. Under unit weights, the schema of the proof has loss three as well, but a plain design attains it by leaving $BDE \to A$, $DF \to C$ and $AF \to C$ unenforced, so the proposition needs the weights. Under unit weights, every random instance we examined has a design with minimum loss whose enforced FDs are generated by Ψ and local weakened FDs, but whether such a design always exists is open.

Tightness of the cover condition. The cover condition of Lem. 5.3 cannot be strengthened to confine Δ to the critical minimal FDs and the FDs that violate their schemata.

Proposition A.5 (Tightness of the cover condition). *There is a schema (R, Σ) such that every BCNF design of (R, Σ) with minimum loss leaves a minimal FD of Σ unenforced that is uncritical and violates the schema of no critical minimal FD of Σ.*

The optimal designs of this schema are not plain, since every plain design leaves at least four minimal FDs unenforced, while the design of the proof leaves three.

Isolated critical schemata. The following proposition identifies a case in which Alg. 3 is optimal.

Proposition A.6 (Isolated critical schemata). *If the critical schemata of $\mathcal{D}_0$ share no attribute with each other or with any other schema of $\mathcal{D}_0$, except for a schema that is a minimal key of (R, Σ), then line 9 of Alg. 3 succeeds and the returned design has loss $\lambda_w(R, \Sigma)$.*

In this case, (R, Σ) is the attribute-disjoint union of the critical schemata and of the remaining attributes, on which the other schemata form a design with loss zero, so the local optima add up to the global optimum by Lem. 4.2, and a key schema only joins the parts.

Branches of Repair. Tab. 7 lists for every FD set of Sec. 6 with positive loss the loss of every design that Alg. 3 examines before it returns the lightest. When the assembled set Γ^+ is enforceable, these are the design of Γ, the uncritical design (column O) of Thm. 3.22 and the ascents of Alg. 4 from both. Otherwise, the fallback examines the uncritical design of (R, Γ^+) in place of the design of Γ, a greedy enforceable subset of Γ^+, the global Drop with the remaining budget, which includes its own ascents, and the ascent from the uncritical design of (R, Γ^+). A dash marks a step that its budget cut or that Drop had already run. The assembled design alone is never the lightest: on abalone and echo the ascent from it recovers the minimal FDs that no schema of iConf embeds, on the six sets of the fallback the time-limited Drop or the ascent from the uncritical design of (R, Γ^+) wins, and on breast the uncritical design of (R, Σ) is optimal as well.

The price of plainness. The following proposition bounds the price of confining the search of MinLoss to plain designs and shows that the bound is attained.

Proposition A.7 (Price of plainness). *Let $\lambda^{\mathrm{p}}_w(R, \Sigma)$ denote the minimum loss of the plain BCNF designs of (R, Σ), and let O be the set of obstructions of (R, Σ). Then $\lambda_w(R, \Sigma) \le \lambda^{\mathrm{p}}_w(R, \Sigma) \le w(O)$. For every $m \ge 2$ there is a schema (R_m, Σ_m) with $m + 3$ attributes such that $\lambda(R_m, \Sigma_m) = 1$ and $\lambda^{\mathrm{p}}(R_m, \Sigma_m) = |O| = \binom{m}{\lfloor m/2 \rfloor}$, and every BCNF design of (R_m, Σ_m) with loss one enforces $\binom{m}{\lfloor m/2 \rfloor}$ weakened forms of the same unenforced minimal FD.*

The upper bound holds since the uncritical design is plain. As $\lambda(R, \Sigma) \ge 1$ whenever $O \ne \emptyset$, plain designs lose at most $|O|$ times the minimum under unit weights, and (R_m, Σ_m) attains this factor. Its FDs are $B \to A$ and, for every set Y of $\lfloor m/2 \rfloor$ attributes among $C_1, \ldots, C_m$, the FDs $BY \to E$ and $AYE \to B$, so that $m = 2$ yields the first schema of Prop. A.3. An optimal design gives up $B \to A$ and enforces $BY \to A$ on every schema $ABYE$, while a plain design gives up $AYE \to B$ or $BY \to E$ for every Y. Hence, the restriction

Table 6: Notation

Symbol	Meaning	Symbol	Meaning
Σ, Σ^+, Σ_a	FD set, its closure, its atomic cover of minimal FDs	X^+_Σ	attribute closure of X under Σ
$(\mathcal{D}, \kappa)$	key design with decomposition $\mathcal{D}$ and declared keys $\kappa(S)$ for every $S \in \mathcal{D}$ (Def. 3.1)	$\mathcal{E}$	enforced FDs of a design, the closure of its key dependencies
λ_w, λ	dependency loss with weights w and with unit weights (Def. 3.10)	$\nu(R, \Sigma)$	number of non-key minimal FDs of Σ (Prop. 3.12)
critical	a schema of a 3NF decomposition that is not in BCNF (Sec. 2), or a minimal FD $X \to A$ of $\mathcal{E}$ such that $(XA, \mathcal{E}[XA])$ is not in BCNF (Def. 3.17)	$\mathcal{U}$, O	uncritical minimal FDs of Σ, obstructions $\Sigma_a - \mathcal{U}^+$ (Def. 3.21)
Ψ	uncritical set, which generates an enforceable FD set (Cor. 3.19)	Δ, W	minimal FDs of Σ left unenforced, weakened FDs enforced instead (Alg. 1)
$C(\Delta)$, $Q(g, \Delta)$	critical minimal FDs whose schemata Δ relieves, requirement set of g (Lem. 5.3, Prop. 5.4)	$\mathcal{D}_0$, Γ	3NF decomposition and key dependencies of the local designs in Repair (Alg. 3)

Table 7: Losses of the Designs that Repair Examines, Whether the Assembled Set Γ^+ Is Enforceable, and the Design It Returns (dash: not run)

Data set	Γ^+ enf.	Γ^+	O	greedy	Ascend(Γ^+)	Ascend(O)	Drop	returned
abalone	yes	22	11	–	8	8	–	Ascend(Γ^+)
breast	yes	5	3	–	3	3	–	Ascend(Γ^+)
bridges	no	30	28	30	11	11	11	Ascend(Γ^+)
echo	yes	150	29	–	11	11	–	Ascend(Γ^+)
ncvoter	no	248	88	241	65	–	48	Drop
weather	no	195	192	166	91	–	84	Drop
lineitem	no	424	145	423	71	–	69	Drop
uniprot	no	856	342	780	444	–	136	Drop
hepatitis	no	2,463	2,007	–	1,361	–	992	Drop

Table 8: Unenforced Minimal FDs of the Eight Key Designs that Span Several Schemata (the Rest of the Loss of Tabs. 2 and 3 Lies Inside One Schema)

Data set	key-only	BCNF	greedy	iConf	classical	Hybrid	Repair	Drop
abalone	0	95	92	2	13	17	1	1
breast	0	38	37	1	1	1	1	2
bridges	0	103	99	7	9	13	2	3
echo	0	463	392	52	39	81	1	1
ncvoter	0	704	668	59	51	248	19	19
weather	0	869	858	54	55	209	13	13
hepatitis	0	8,211	7,995	961	1,187	2,230	550	–
food	0	4	4	4	4	–	1	1
price paid	0	2	1	0	0	–	1	1
hospital	0	4	1	4	4	–	1	1
TPC-DS	0	180	51	0	0	–	0	0

of Drop to plain designs in Thm. 5.5 can lose a factor exponential in the number of attributes, although at most linear in the size of the atomic cover, as $O \subseteq \Sigma_a$.

Realizability and upper bounds. A set of unenforced minimal FDs is no certificate of membership in k-loss BCNF that can be checked in polynomial time either, even when the atomic cover is given.

Theorem A.8 (Realizability). *Given (R, Σ) with $\Sigma = \Sigma_a$ and a set $\Delta \subseteq \Sigma_a$, deciding whether some BCNF design of (R, Σ) leaves exactly Δ unenforced is coNP-hard, even when $|\Delta| = 1$, and so is deciding whether some plain BCNF design (Def. 5.1) of (R, Σ) does.*

The proof reduces from the unsatisfiability of a CNF formula. The literal attributes together with an attribute T determine the attribute C_j of every clause that they satisfy, the clause attributes determine an attribute Z, and the FD $T \to Z$ keeps the atomic cover small. Once $T \to Z$ is unenforced, the schema of the FD from the literal attributes and Z to T exhibits a non-key FD exactly when a consistent set of literals satisfies every clause. Hence, the tests in line 9 of Alg. 1 and in line 7 of Alg. 2 are coNP-hard, which is consistent with their running time polynomial in the size of the atomic cover of $\mathcal{E}$, since this atomic cover may be exponentially larger than Σ_a. The following theorem gives upper bounds.

Theorem A.9 (Upper bounds). *Deciding whether $\lambda_w(R, \Sigma) \le k$ for a given schema (R, Σ), weights w and number k is in NEXPTIME. When $\Sigma = \Sigma_a$, deciding whether some plain BCNF design (Def. 5.1) of (R, Σ) has loss at most k is in Σ^p_2.*

For plain designs, a certificate consists of the set Δ and of at most $|\Sigma_a| \cdot |R|$ uncritical minimal FDs of $(\Sigma_a - \Delta)^+$ that imply $\Sigma_a - \Delta$, and a coNP oracle verifies that they are uncritical. With the atomic cover given, MinLoss for plain designs is therefore NP-hard by Thm. 4.3, whose optimal designs are plain, and in Σ^p_2, and its inner test is coNP-hard by Thm. A.8. For unrestricted designs, the same certificate must include the weakened FDs, and membership in Σ^p_2 follows if every enforceable FD set has an equivalent one that is generated by polynomially many uncritical minimal FDs. Prop. A.7 shows that a number of weakened forms linear in $|\Sigma_a|$ may be necessary, and whether polynomially many always suffice is open.

Repair distance. The *repair distance* $\delta_\Sigma(\mathcal{E}) = \min\{|\mathcal{F}| \mid \mathcal{F} \subseteq \Sigma^+,\ (\mathcal{E} \cup \mathcal{F})^+ = \Sigma^+\}$ satisfies (R1) but violates (R2), since it charges the unenforced rules by the effort of restoring them rather than by the rules at risk.

Proposition A.10 (Repair distance). *For every key design with enforced FDs $\mathcal{E}$, $\delta_\Sigma(\mathcal{E}) \le \lambda(\mathcal{D}, \kappa)$. For every $n \ge 1$ there is a schema with a BCNF design such that $\delta_\Sigma(\mathcal{E}) = 1$ and $\lambda(\mathcal{D}, \kappa) = n + 1$, and the join of some state that satisfies the keys violates each of the $n + 1$ unenforced minimal FDs.*

Decomposition by sources.

Prop. A.6 and Lem. 4.2 require parts of a schema that share no attribute. Real schemata share attributes that nothing determines, such as an identifier, and the following result shows that such attributes do not couple the parts. We call an attribute a *source* of (R, Σ) if it is the right-hand side of no minimal FD of Σ, so that $X \to A \in \Sigma^+$ implies $A \in X$ for every source A. The *parts* of (R, Σ) are the connected components of the graph on the non-sources that joins B and A whenever $B \in X$ for some minimal FD $X \to A$ of Σ. For a part P, let R_P consist of P and the sources in the left-hand

sides of the minimal FDs with right-hand side in P, and let Σ_P be the set of these minimal FDs, which are FDs over R_P. A BCNF design with enforced FDs $\mathcal{E}$ is *part-respecting* if every minimal FD $X \to A$ of $\mathcal{E}$ satisfies $XA \subseteq R_P$ for the part P that contains A. Every plain design is part-respecting, since its enforced FDs are generated by minimal FDs of Σ.

Theorem A.11 (Decomposition by sources). *Let $P_1, \dots, P_r$ be the parts of (R, Σ), and let $R_i = R_{P_i}$ and $\Sigma_i = \Sigma_{P_i}$. Then $\lambda_w(R, \Sigma) \leq \sum_{i=1}^{r} \lambda_w(R_i, \Sigma_i)$, and the sum is the minimum loss of the part-respecting BCNF designs of (R, Σ). Moreover, $\lambda_w^{\mathrm{p}}(R, \Sigma) = \sum_{i=1}^{r} \lambda_w^{\mathrm{p}}(R_i, \Sigma_i)$.*

Gluing designs for the parts yields a part-respecting design whose loss is the sum, and conversely, the enforced FDs of a part-respecting design restrict to an enforceable FD set of every part. Lem. 4.2 is the case without shared sources, and the attribute-disjoint schemata in Prop. A.6 are unions of parts and sources. For example, customers with a billing and a shipping address, $C \to S_1T_1$, $S_1T_1 \to Z_1$, $Z_1 \to T_1$, $C \to S_2T_2$, $S_2T_2 \to Z_2$ and $Z_2 \to T_2$, have the source C and the two parts $S_1T_1Z_1$ and $S_2T_2Z_2$, each with loss one, so the loss of the schema is two by Thm. A.11 and Prop. 5.4, and each address is designed on its own.

Corollary A.12 (Parameterized tractability). *Given $\Sigma = \Sigma_a$, the minimum loss of the plain BCNF designs and the minimum loss of the part-respecting BCNF designs of (R, Σ), together with designs that attain them, can be computed in time $f(m) \cdot (|R| \cdot |\Sigma_a|)^{O(1)}$, where m is the largest number of attributes of a schema R_i and f is a computable function.*

Indeed, the parts arise from a graph search over Σ_a, and Drop, restricted to plain designs or not, solves every part (R_i, Σ_i) in time bounded by a function of $|R_i|$. In the example above, the parts have three attributes plus the source C, however many addresses a customer has. We conjecture that the restriction to part-respecting designs loses nothing, that is, that equality holds in Thm. A.11. The difficulty is that a minimal FD of an enforceable FD set may combine attributes of several parts, and its restriction to a part need not be enforceable.

Axioms of the loss. The requirements (R1) to (R3) of Sec. 3 read formally as follows. A *loss measure* μ assigns to every FD set Σ over R and every closed FD set $\mathcal{E} \subseteq \Sigma^+$ a value $\mu_\Sigma(\mathcal{E}) \geq 0$. Let $F(\Sigma)$ denote the set of non-trivial FDs $X \to A$ in Σ^+ with a single attribute A.

(R1) *Representation independence:* $\mu_\Sigma = \mu_{\Sigma'}$ whenever $\Sigma^+ = \Sigma'^+$.

(R2) *Rule additivity:* there is a cost function $c_\Sigma : F(\Sigma) \to \mathbb{R}_{\geq 0}$ with $\mu_\Sigma(\mathcal{E}) = \sum_{\varphi \in F(\Sigma) - \mathcal{E}} c_\Sigma(\varphi)$ for every $\mathcal{E}$, that is, every FD that a design leaves unenforced is a rule at risk (Prop. 3.9), and the costs of these rules add up.

(R3) *No double counting:* $c_\Sigma(X \to A) = 0$ whenever $Y \to A \in \Sigma^+$ for some $Y \subset X$.

Requirement (R3) holds for every cost that measures violations, since a relation that violates $X \to A$ also violates $Y \to A$ for every $Y \subset X$.

Proposition A.13 (Axioms). *A loss measure satisfies (R1) to (R3) iff $\mu_\Sigma(\mathcal{E}) = \sum_{\varphi \in \Sigma_a - \mathcal{E}} w(\varphi)$ for weights $w : \Sigma_a \to \mathbb{R}_{\geq 0}$ that depend only on Σ^+. It satisfies in addition $c_\Sigma(\varphi) = 1$ for every minimal FD φ of Σ iff $\mu_\Sigma(\mathcal{E}) = |\Sigma_a - \mathcal{E}|$.*

Algorithm 4 Ascend$(R, \Sigma, w, \mathcal{E})$

Require: Relation schema R, FD set Σ over R, weights w on Σ_a, an FD set $\mathcal{E} \subseteq \Sigma^+$ that is enforceable for (R, Σ), bounds s and k, a time budget
Ensure: A BCNF design of (R, Σ) with loss at most $w(\Sigma_a - \mathcal{E})$

```
1:  Ψ ← the uncritical minimal FDs of E, and Ψ* ← Ψ                ▷ Ψ⁺ = E
2:  repeat
3:      repeat                                                    ▷ climb
4:          for all non-tabu g = Z → B ∈ Σ_a − Ψ⁺ in decreasing order of w(g) do
5:              if Ψ ∪ {g} is uncritical then
6:                  Ψ ← Ψ ∪ {g}
7:              else
8:                  V ← the X → A ∈ Ψ with XA ⊆ ZB and X not a key of ZB
                    under Ψ ∪ {g}                                 ▷ the violators of the schema of g
9:                  if Ψ' = (Ψ − V) ∪ {g} is uncritical and loses no more than Ψ then
10:                     Ψ ← Ψ'                                    ▷ at equal loss at most s times in a row
11:     until Ψ is unchanged
12:     Ψ* ← Ψ if it loses no more than Ψ*, and Ψ ← a random uncritical kick of Ψ*
13: until k kicks in a row leave w(Σ_a − Ψ*⁺) unchanged, or the budget is exhausted
14: return the design of Cor. 3.19 for Ψ*
```

Goal-directed weakening. The implementation of line 5 of Alg. 1 grows W by a goal-directed search, which the following lemma makes complete.

Lemma A.14 (Goal-directed weakening). *Let $\Delta \subseteq \Sigma_a$, let $\mathcal{E} \subseteq \Sigma^+$ be a closed FD set that contains $\Sigma_a - \Delta$ and no element of Δ, and let $g = Y \to B$ be a critical minimal FD of $\mathcal{E}$ that the uncritical minimal FDs of $\mathcal{E}$ do not imply. Then every enforceable FD set $\mathcal{E}' \supseteq \mathcal{E}$ with $\mathcal{E}' \cap \Delta = \emptyset$ contains a minimal FD $X' \to A \notin \mathcal{E}$ of $\mathcal{E}'$ such that $X \subset X' \subseteq Y^+_{\mathcal{E}}$ for some $X \to A \in \Delta$.*

Hence, the search picks such an FD g and branches only over the weakenings $X' \to A \notin \mathcal{E}$ of the elements $X \to A$ of Δ with $X' \subseteq Y^+_{\mathcal{E}}$. Neither the enumeration of W nor the cover condition can be confined further (Prop. A.3, Prop. A.4 and Prop. A.5 in Appendix A).

The algorithms on the running example.

Example A.15 (Drop). For (R, Σ) from Ex. 1.1, the budget $b = 0$ fails since Σ^+ is not enforceable by Ex. 3.20. At budget $b = 1$, line 3 discards the sets $\{C \to S\}$, $\{C \to T\}$ and $\{C \to Z\}$, since none of their elements violates STZ and the uncritical minimal FDs $C \to S$, $C \to T$, $C \to Z$ and $Z \to T$ do not imply $ST \to Z$. The two remaining sets $\{ST \to Z\}$ and $\{Z \to T\}$ both succeed with $W = \emptyset$, and under unit weights the order of enumeration decides which of them is examined first. The set $\Delta = \{ST \to Z\}$ leaves $\Psi = \{C \to S, C \to T, C \to Z, Z \to T\}$, whose decomposition $\{CSZ, ZT\}$ is in BCNF, and the algorithm returns $\mathcal{D}_2$. The set $\Delta = \{Z \to T\}$ yields the plain design $\mathcal{D}'_4$, which is examined first whenever the weight of $Z \to T$ is smaller than that of $ST \to Z$. □

The weights thus decide between designs of equal loss.

Example A.16 (Ascend). For (R, Σ) from Ex. 1.1 and $\mathcal{E} = \mathcal{U}^+$, the algorithm starts from $\Psi = \{C \to S, C \to T, C \to Z, Z \to T\}$ with the single candidate $ST \to Z$. Its addition fails since $Z \to T$ violates STZ, and its exchange for $Z \to T$ keeps the loss at one under unit weights, so the climb accepts it and returns a design with the enforced FDs of the plain design $\mathcal{D}'_4$ of Ex. A.15. Had $Z \to T$ weighed more, it would reject the exchange and return the uncritical design, which is then optimal. □

Pseudocode of Ascend and the loss–time trade-off. Alg. 4 and Fig. 3 complement Sec. 5 and Sec. 6.3.

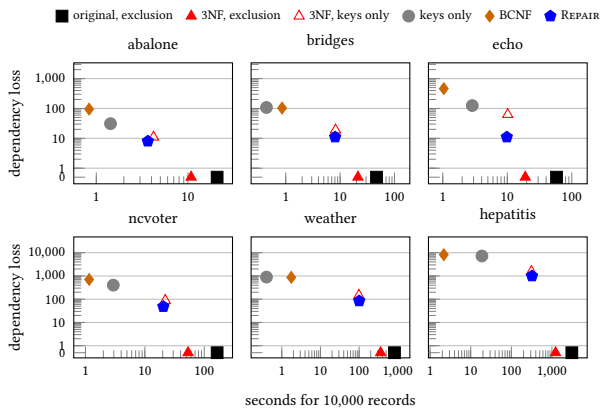


Figure 3: Time for Inserting 10,000 Records against the Dependency Loss of the Designs

B PROOFS

Proof of Lemma 3.3. Since $\mathcal{E} \subseteq \Sigma^+$, the FD $X \to A$ is in Σ^+ and $A \notin X$. Hence, there is a minimal $Y \subseteq X$ with $Y \to A \in \Sigma^+$, which is a minimal FD of Σ. □

Proof of Lemma 3.8. Dependency preservation holds since $\mathcal{E}$ is the closure of key dependencies, each of which is embedded in some schema of $\mathcal{D}$. If $(\mathcal{D}, \kappa)$ is locally enforced, then every non-trivial FD $X \to Y \in \mathcal{E}[S]$ has a left-hand side that contains some $K \in \kappa(S)$, so X is a key of S for $\mathcal{E}[S]$. For the converse, let $X \to Y \in \mathcal{E}[S]$ be non-trivial. Since $(S, \mathcal{E}[S])$ is in BCNF, X is a key of S for $\mathcal{E}[S]$, so X contains a minimal key of $(S, \mathcal{E}[S])$, which belongs to $\kappa(S)$ by assumption. □

Proof of Proposition 3.9. For statement 1, local enforcement gives $\mathcal{E}[S] = \{K \to S \mid K \in \kappa(S)\}^+[S]$, and r_S satisfies these key dependencies, so it satisfies $\mathcal{E}[S]$. By Lem. 3.8, $(S, \mathcal{E}[S])$ is in BCNF, and no relation over a BCNF schema that satisfies its FDs contains a redundant data value [62]. For statement 2, let t_1 and t_2 be tuples of the join that agree on X, and let $X \to Y \in \mathcal{E}$. The closure $X^+_{\mathcal{E}}$ arises from X by repeatedly adding a schema $S \in \mathcal{D}$ that has a key $K \in \kappa(S)$ within the attributes reached so far. If t_1 and t_2 agree on these attributes, then $t_1[S]$ and $t_2[S]$ are tuples of r_S that agree on its key K, so t_1 and t_2 agree on S as well. By induction, they agree on $X^+_{\mathcal{E}} \supseteq Y$. Statement 3 is the definition of soundness. For the last statement, the join of a state that satisfies the keys satisfies every FD of $\mathcal{E}$ by statement 2. Conversely, if $\varphi \notin \mathcal{E}$, then some relation r over R satisfies $\mathcal{E}$ and violates φ, since $\mathcal{E}$ is closed. Its projections form a state that satisfies the keys, and their join contains r, so it violates φ. □

Proof of Proposition A.13. The minimal FDs of Σ are the elements of $F(\Sigma)$ whose left-hand side has no proper subset that determines the right-hand side, so (R3) leaves non-zero costs only on Σ_a, and (R2) becomes $\mu_\Sigma(\mathcal{E}) = \sum_{\varphi \in \Sigma_a - \mathcal{E}} c_\Sigma(\varphi)$. By (R1) and since Σ_a is determined by Σ^+, the weights $w = c_\Sigma$ restricted to Σ_a depend only on Σ^+. Conversely, the measure $\sum_{\varphi \in \Sigma_a - \mathcal{E}} w(\varphi)$ satisfies (R2) with $c_\Sigma = w$ on Σ_a and $c_\Sigma = 0$ elsewhere, (R3) by construction, and (R1) since Σ_a and w depend only on Σ^+. The second statement is the case $w = 1$. □

Proof of Proposition A.10. The set $\Sigma_a - \mathcal{E}$ restores Σ^+, since Σ_a is a cover of Σ, so $\delta_\Sigma(\mathcal{E}) \leq |\Sigma_a - \mathcal{E}|$. For the second statement, take the schema (R_n, Σ_n) of the proof of Prop. 3.16 and a BCNF design of the classical algorithm, which enforces $\mathcal{E} = \{Z \to TP_1 \cdots P_n\}^+$ and has loss $n + 1$, as $ST \to Z$ and every $ST \to P_i$ are unenforced. Adding $ST \to Z$ restores Σ^+_n, and $\mathcal{E} \neq \Sigma^+_n$, so $\delta_{\Sigma_n}(\mathcal{E}) = 1$. The last claim is the last statement of Prop. 3.9. □

Proof of Proposition 3.13. A relation over R satisfies a declared key dependency $K \to S$ iff its projection onto S does, and it satisfies all of them iff it satisfies their closure $\mathcal{E}$. If r' violates Σ, then it violates some minimal FD of Σ, since Σ_a is a cover of Σ. If, in addition, r' satisfies $\mathcal{E}$, then every minimal FD of Σ that r' violates lies outside $\mathcal{E}$, so the event is contained in the union of the events that r' violates φ for $\varphi \in \Sigma_a - \mathcal{E}$, whose probability is at most $\sum_{\varphi \in \Sigma_a - \mathcal{E}} p(\varphi) = \lambda_p(\mathcal{D}, \kappa)$. The expected number of violated minimal FDs outside $\mathcal{E}$ is the same sum by linearity of expectation. For a plain design, $\mathcal{E}$ is generated by $\Sigma_a \cap \mathcal{E}$, so a relation that violates $\mathcal{E}$ violates a minimal FD of Σ in $\mathcal{E}$. If, in addition, r' never violates two minimal FDs of Σ, then the events for $\varphi \in \Sigma_a - \mathcal{E}$ are disjoint, and an r' that violates such a φ satisfies $\mathcal{E}$ and violates Σ, so the union bound is an equality. □

Proof of Theorem 3.24. A relation over R satisfies $\mathcal{E}$ iff it satisfies the declared key dependencies, so global consistency asks for a relation r that satisfies them with $r_S \subseteq r[S]$ for all S.

Necessity. Let $S \neq T$, $K \in \kappa(T)$ with $K \subseteq S^+_{\setminus T}$, and $B \in (T - K) \cap S^+_{\setminus T}$. Let every relation r_U with $U \neq T$ consist of the tuple that is 0 on U, and let r_T consist of the tuple that is 1 on B and 0 elsewhere on T. Every relation has one tuple, so the state satisfies the keys. Suppose r satisfies the key dependencies and contains a tuple u with $u[S] = 0$. Induction along the closure computation of $S^+_{\setminus T}$ shows $u[S^+_{\setminus T}] = 0$: whenever u is 0 on a key K' of $U \neq T$, it agrees with the tuple of r_U on K' and hence on U. Then $u[K] = 0$ agrees with the tuple of r_T on K, so $u[B] = 1$, which contradicts $u[B] = 0$. Hence, the state is not globally consistent.

Sufficiency. Suppose the condition holds, and let a state satisfy the keys. For a stored tuple $t \in r_S$, we extend t to a partial tuple $e(t)$ by lookups. Initially, $e(t) = t$ and only S is *used*. While some schema T that is not used has a key $K \subseteq \mathrm{dom}(e(t))$, mark T as used, and for every $B \in T - K$ set $e(t)[B]$ to $s[B]$ if r_T contains a tuple s with $s[K] = e(t)[K]$, and to the term $\nu_{T,K,B}(e(t)[K])$ otherwise. Since only schemata other than T were used before, $\mathrm{dom}(e(t)) \subseteq S^+_{\setminus T}$, so $T - K$ is disjoint from $\mathrm{dom}(e(t))$ by the condition, and no value is overwritten. If two keys K, K' of T lie in $\mathrm{dom}(e(t))$, then $T \cap \mathrm{dom}(e(t)) \subseteq K \cap K'$ by the condition, so $K = K'$, and the lookup is determined. Finally, $e(t)$ is completed by terms $\nu_{t,B}$ that are unique to t and B, and $r = \{e(t) \mid t \text{ stored}\}$ satisfies $r_S \subseteq r[S]$ for all S, where terms are read as fresh constants. A term $\nu_{T,K,B}(\cdot)$ arises only from a lookup in T by K, which happens at most once in every extension.

It remains to show that r satisfies every declared key $K \to T$. Let $e(t_1)[K] = e(t_2)[K]$. No attribute of K carries a term $\nu_{t,B}$, unless $t_1 = t_2$, so $K \subseteq \mathrm{dom}(e(t_i))$ before completion, and T is used in

both extensions, either as the schema of t_i or by a lookup with some key K_i; in both cases $T \subseteq \mathrm{dom}(e(t_i))$. We say that $e(t_i)$ *hits* $s \in r_T$ if $t_i = s$ or the lookup in T found s; then $e(t_i)[T] = s$. If both hit tuples s_1, s_2, then $s_1[K] = s_2[K]$, so $s_1 = s_2$ by the key K of r_T, and $e(t_1)[T] = e(t_2)[T]$. If $e(t_1)$ hits s_1 and $e(t_2)$ hits no tuple, then $e(t_2)$ carries terms $v_{T,K_2,\cdot}$ on $T - K_2$, so $K \subseteq K_2$, since $e(t_1)[K] = s_1[K]$ consists of constants; as K and K_2 lie in $\mathrm{dom}(e(t_2))$ when T is looked up, $K = K_2$, and s_1 would have been found, a contradiction. If neither hits a tuple, then the terms on $K - K_1$ and $K - K_2$ arise from the lookups in T, and the values of K known before these lookups contain no term $v_{T,\cdot,\cdot}$. Hence, $K - K_1$ is empty iff $K - K_2$ is, and in this case $K = K_1 = K_2$ as before; otherwise the terms on $K - K_1 = K - K_2$ coincide, which forces $K_1 = K_2$ and $e(t_1)[K_1] = e(t_2)[K_2]$. In both cases the lookups produce the same values on T, so $e(t_1)[T] = e(t_2)[T]$. Thus, r witnesses global consistency. The condition is tested with $O(|\mathcal{D}|^2)$ closure computations. □

Proof of Proposition 3.12. The enforced FDs of $(\{R\}, \kappa)$ are the closure $\mathcal{E}$ of the key dependencies of Σ. For every $X \subseteq R$, $X^+_{\mathcal{E}} = R$ if X contains a minimal key and $X^+_{\mathcal{E}} = X$ otherwise, so every non-trivial FD in $\mathcal{E}$ has a key as left-hand side. Hence, the design is locally enforced, it is sound as $\mathcal{D} = \{R\}$, and $\Sigma_a - \mathcal{E}$ consists of the non-key minimal FDs. For the second statement, let $X \to A$ be a non-key minimal FD of Σ and suppose $X \to A \in \mathcal{E}'$ for the enforced FDs $\mathcal{E}'$ of $(\{R\}, \kappa')$. Local enforcement requires that X contains some $K \in \kappa'(R)$, so $X \to R \in \Sigma^+$, which contradicts that X is not a key. □

Proof of Theorem 3.15. Let $(\mathcal{D}, \kappa)$ be a BCNF design with $\lambda(\mathcal{D}, \kappa) = 0$ and enforced FDs $\mathcal{E}$. Then $\Sigma_a \subseteq \mathcal{E} \subseteq \Sigma^+$, so $\mathcal{E} = \Sigma^+$ as Σ_a is a cover of Σ. Then $\mathcal{D}$ is lossless for Σ by soundness, and dependency-preserving and in BCNF for Σ by Lem. 3.8. Conversely, let $\mathcal{D}$ be a lossless, dependency-preserving decomposition of (R, Σ) into BCNF, and let $\kappa(S)$ consist of the minimal keys of $(S, \Sigma[S])$ for every $S \in \mathcal{D}$. Since $(S, \Sigma[S])$ is in BCNF, every FD in $\Sigma[S]$ is implied by the key dependencies of S, so $\mathcal{E}(\mathcal{D}, \kappa)$ is the closure of $\bigcup_{S \in \mathcal{D}} \Sigma[S]$, which is Σ^+ by dependency preservation. Hence, $(\mathcal{D}, \kappa)$ is sound, locally enforced, and has loss zero. □

Proof of Proposition 3.16. Let $R_n = \{S, T, Z, P_1, \ldots, P_n\}$ and $\Sigma_n = \{ST \to Z, Z \to T, Z \to P_1, \ldots, Z \to P_n\}$. The minimal keys are ST and SZ, and the minimal FDs are $ST \to Z$, $ST \to P_i$, $Z \to T$ and $Z \to P_i$ for $i = 1, \ldots, n$. The non-key minimal FDs are $Z \to T$ and $Z \to P_i$, so the design $(\{R_n\}, \kappa)$ has loss $n+1$. A set that contains neither Z nor ST determines only itself, and a set that contains SZ or ST is a key, so every violating FD $X \to W$ of a schema has $Z \in X$ and $S \notin X$. A step of the classical algorithm on such an FD splits a schema S' into XW and $S' - W$, which both contain Z, and only $S' - W$ contains S, since no set without S determines S. As long as the schema that contains S also contains some attribute B of $TP_1 \cdots P_n$, the FD $Z \to B$ violates it, since Z does not determine S. Hence, every run ends with the schema SZ with key SZ and schemata that contain Z but not S, whose only minimal key is Z. Their keys imply $Z \to T$ and every $Z \to P_i$, while the closure of ST under them is ST, so they imply neither $ST \to Z$ nor any $ST \to P_i$, and the loss is $n + 1$. The design $\{STZ, ZP_1 \cdots P_n\}$ with keys ST and SZ of STZ and key Z of $ZP_1 \cdots P_n$ is sound since $STZ \cap ZP_1 \cdots P_n = Z$ determines $P_1 \cdots P_n$, it is locally enforced, and it enforces every minimal FD except $Z \to T$. Finally, $ST \to Z$ is the only minimal FD of Σ_n with right-hand side Z, and it is critical since $Z \to T$ is a non-key FD of $(STZ, \Sigma_n[STZ])$, so Σ^+_n is not enforceable by Thm. 3.18 and $\lambda(R_n, \Sigma_n) \geq 1$. □

Proof of Theorem 3.18. Statement 1 implies statement 2 by Lem. 3.8 and soundness. For the converse, let $\mathcal{D}$ be a lossless, dependency-preserving decomposition of $(R, \mathcal{E})$ into BCNF, and let $\kappa(S)$ consist of the minimal keys of $(S, \mathcal{E}[S])$ for every $S \in \mathcal{D}$. As in the proof of Thm. 3.15, $(\mathcal{D}, \kappa)$ is a key design of (R, Σ) with $\mathcal{E}(\mathcal{D}, \kappa) = \mathcal{E}$, it is sound since $\mathcal{D}$ is lossless for $\mathcal{E}$, and it is locally enforced by Lem. 3.8. Next, let $\mathcal{D}$ be as before, let $S \in \mathcal{D}$, let K be a minimal key of $(S, \mathcal{E}[S])$, and let $A \in S - K$. Some minimal $X \subseteq K$ satisfies $X \to A \in \mathcal{E}$. Every non-trivial FD $Y \to B \in \mathcal{E}[XA]$ belongs to $\mathcal{E}[S]$, so Y is a key of S for $\mathcal{E}$ since $(S, \mathcal{E}[S])$ is in BCNF, and therefore Y is a key of XA for $\mathcal{E}$. Hence, $X \to A$ is uncritical. The FDs obtained in this way imply every key dependency $K \to S$ of $\mathcal{D}$, and these imply $\mathcal{E}$ since $\mathcal{D}$ is dependency-preserving and in BCNF for $\mathcal{E}$. Finally, let the uncritical minimal FDs U of $\mathcal{E}$ form a cover of $\mathcal{E}$, and let K be a minimal key of $(R, \mathcal{E})$. The decomposition $\{XA \mid X \to A \in U\} \cup \{K\}$ is in BCNF for $\mathcal{E}$, since $(K, \mathcal{E}[K])$ has no non-trivial FD by the minimality of K. It is dependency-preserving for $\mathcal{E}$ since U is a cover of $\mathcal{E}$, and it is lossless for $\mathcal{E}$ since it is dependency-preserving and contains a key [13]. □

Proof of Corollary 3.19. Let $\mathcal{E} = \Psi^+$, let K be a minimal key of $(R, \mathcal{E})$, and let $\mathcal{D}$ consist of the schemata XA of the elements $X \to A$ of Ψ and of K. Every $(XA, \mathcal{E}[XA])$ is in BCNF since Ψ is uncritical, and $(K, \mathcal{E}[K])$ has no non-trivial FD by the minimality of K. Hence, $\mathcal{D}$ is in BCNF for $\mathcal{E}$, it is dependency-preserving for $\mathcal{E}$ since Ψ is a cover of $\mathcal{E}$ whose elements are embedded in its schemata, and it is lossless for $\mathcal{E}$ since it contains a key [13]. As in the proof of Thm. 3.18, declaring the minimal keys of $(S, \mathcal{E}[S])$ on every $S \in \mathcal{D}$ yields a BCNF design of (R, Σ) with enforced FDs $\mathcal{E}$, so $\mathcal{E}$ is enforceable. □

Proof of Lemma A.2. Let $X \to A$ be an uncritical minimal FD of Σ, and let $\mathcal{U} \subseteq \mathcal{E} \subseteq \Sigma^+$ with $\mathcal{E}$ closed. Since $\mathcal{E} \subseteq \Sigma^+$ contains $X \to A$, the FD $X \to A$ is a minimal FD of $\mathcal{E}$. Let $Y \to B \in \mathcal{E}[XA]$ be non-trivial, and let $Z \in XA - Y$. Since $Y \to B \in \Sigma^+[XA]$ and $X \to A$ is uncritical for Σ^+, Y is a key of XA for Σ, so some minimal FD $Y' \to Z$ of Σ satisfies $Y' \subseteq Y$. If $Y' \to Z$ were critical for Σ^+, some non-trivial FD $V \to W \in \Sigma^+[Y'Z]$ would have a left-hand side V that is not a key of $Y'Z$ for Σ, but $V \to W \in \Sigma^+[XA]$ since $Y'Z \subseteq XA$, so V is a key of XA for Σ and hence of $Y'Z$, a contradiction. Consequently, $Y' \to Z \in \mathcal{U} \subseteq \mathcal{E}$ and $Y \to Z \in \mathcal{E}$ for every $Z \in XA - Y$, so Y is a key of XA for $\mathcal{E}$. Hence, $(XA, \mathcal{E}[XA])$ is in BCNF, that is, $X \to A$ is uncritical for $\mathcal{E}$. □

Proof of Theorem 3.22. Let $\mathcal{E} = \mathcal{U}^+$. By Lem. A.2, every element of $\mathcal{U}$ is an uncritical minimal FD of $\mathcal{E}$, and $\mathcal{U}$ is a cover of $\mathcal{E}$. Hence, the uncritical minimal FDs of $\mathcal{E}$ form a cover of $\mathcal{E}$, and $\mathcal{E}$ is enforceable for (R, Σ) by Thm. 3.18. A design with enforced FDs $\mathcal{E}$ has loss $w(\Sigma_a - \mathcal{E}) = w(O)$, so $\lambda_w(R, \Sigma) \leq w(O)$. □

Proof of Proposition A.1. Ex. 3.5 provides the schema and the decomposition $\mathcal{D}_3$. □

Proof of Proposition 3.23. The key dependency $K \to K$ is trivial, so the enforced FDs remain $\mathcal{E}$, and every $S \in \mathcal{D}$ remains locally enforced. The projection $\mathcal{E}[K]$ contains no non-trivial FD, since an FD $Y \to B \in \mathcal{E}$ with $YB \subseteq K$ and $B \notin Y$ would make $K - B$ a key of $(R, \mathcal{E})$, so the design is locally enforced on K as well. It is sound since $\mathcal{D} \cup \{K\}$ is dependency-preserving for $\mathcal{E}$, whose generating key dependencies are embedded in its schemata, and contains the key K of $(R, \mathcal{E})$ [13]. For the second statement, let $\mathcal{E}$ be the closure of $\bigcup_{S \in \mathcal{D}} \Sigma[S]$. Then $\Sigma[S] \subseteq \mathcal{E} \subseteq \Sigma^+$ gives $\mathcal{E}[S] = \Sigma[S]$, which is in BCNF and therefore implied by the key dependencies of the minimal keys of $(S, \Sigma[S])$. Hence, declaring these keys yields a locally enforced key design with enforced FDs $\mathcal{E}$, and the first statement applies. □

Proof of Theorem 4.1. For $k = 0$ the statement is Thm. 3.15 together with the NP-hardness of deciding the existence of a lossless, dependency-preserving decomposition into BCNF [6]. This NP-hardness persists under the assumption of Sec. 2, since deleting the set C of constant attributes from R and from the FDs of Σ takes polynomial time and preserves the existence of such a decomposition. Indeed, a schema that contains an attribute of C and an attribute outside C violates BCNF by $\emptyset \to C$, so the schemata disjoint from C form such a decomposition of $R - C$, and conversely every such decomposition of $R - C$ extends by the schema C. For $k > 0$ we reduce from the case $k = 0$. Let (R, Σ) be an instance, and let $(R_1, \Sigma_1), \ldots, (R_k, \Sigma_k)$ be k copies of the schema $(\{S, T, Z\}, \{ST \to Z, Z \to T\})$ over pairwise disjoint sets of fresh attributes. Each copy has loss one, since its key-only design leaves only $Z \to T$ unenforced (Prop. 3.12), while its closure is not enforceable, as $ST \to Z$ is critical and the only minimal FD with right-hand side Z, and the copies share no attribute with R and introduce no constant attribute, so the union satisfies the assumption of Sec. 2. By Lem. 4.2, applied k times, $\lambda(R \cup R_1 \cup \cdots \cup R_k, \Sigma \cup \Sigma_1 \cup \cdots \cup \Sigma_k) = \lambda(R, \Sigma) + k$, so (R, Σ) is in 0-loss BCNF iff the union is in k-loss BCNF. □

Proof of Lemma 4.2. Let $R = R_1 \cup R_2$ and $\Sigma = \Sigma_1 \cup \Sigma_2$. For $X \subseteq R$, the attribute closure X^+_Σ is the union of $(X \cap R_1)^+_{\Sigma_1}$ and $(X \cap R_2)^+_{\Sigma_2}$. Hence, the minimal FDs of Σ are those of Σ_1 together with those of Σ_2, and the closure of a set $X \subseteq R_i$ under any subset of Σ^+ is contained in R_i, since an FD $Y \to B \in \Sigma^+$ with $Y \subseteq R_i$ has $B \in Y^+_\Sigma = Y^+_{\Sigma_i} \subseteq R_i$, as $\emptyset^+_\Sigma = \emptyset$ by the assumption of Sec. 2. We use the characterization of enforceable FD sets in Thm. 3.18. Let $\mathcal{E}$ be enforceable for (R, Σ), and let $\mathcal{E}_i = \mathcal{E}[R_i]$ for $i = 1, 2$. Then $\mathcal{E}_i$ is a closed subset of Σ_i^+, its minimal FDs are the minimal FDs of $\mathcal{E}$ over R_i, and such an FD $X \to A$ is critical for $\mathcal{E}_i$ iff it is critical for $\mathcal{E}$, since both properties depend on $\mathcal{E}[XA] = \mathcal{E}_i[XA]$ only. Every FD $X \to A \in \mathcal{E}_i$ is implied by the uncritical minimal FDs of $\mathcal{E}$, and its derivation uses FDs over R_i only, since the closure of X stays within R_i. Hence, $\mathcal{E}_i$ is enforceable for (R_i, Σ_i), and $\lambda(R_1, \Sigma_1) + \lambda(R_2, \Sigma_2) \leq |(\Sigma_1)_a - \mathcal{E}_1| + |(\Sigma_2)_a - \mathcal{E}_2| = |\Sigma_a - \mathcal{E}|$. Conversely, let $\mathcal{E}_i$ be enforceable for (R_i, Σ_i) with uncritical minimal FDs U_i, and let $\mathcal{E} = (\mathcal{E}_1 \cup \mathcal{E}_2)^+$. The minimal FDs of $\mathcal{E}$ are those of $\mathcal{E}_1$ together with those of $\mathcal{E}_2$, each of them is critical for $\mathcal{E}$ iff it is critical for its $\mathcal{E}_i$, and $U_1 \cup U_2$ is a cover of $\mathcal{E}$. Hence, $\mathcal{E}$ is enforceable for (R, Σ) with $|\Sigma_a - \mathcal{E}| = |(\Sigma_1)_a - \mathcal{E}_1| + |(\Sigma_2)_a - \mathcal{E}_2|$. □

Proof of Corollary 4.4. Let $\mathcal{A}$ be a polynomial-time algorithm whose designs have loss at most $\rho \cdot \lambda(R, \Sigma)$. The design returned for (R, Σ) has loss zero iff (R, Σ) is in 0-loss BCNF, and it has loss zero iff its declared key dependencies imply every FD of Σ, which attribute closures decide in polynomial time. Hence, $\mathcal{A}$ decides 0-loss BCNF, which is NP-hard by Thm. 4.1. Now let $\mathcal{A}$ be a polynomial-time algorithm whose designs have loss at most $\lambda(R, \Sigma) + c$. Given (R, Σ), let (R', Σ') consist of $c + 1$ copies of (R, Σ) over pairwise disjoint sets of attributes, and let $\mathcal{E}$ be the enforced FDs of the design that $\mathcal{A}$ returns for (R', Σ'). If (R, Σ) is in 0-loss BCNF, then $\lambda(R', \Sigma') = 0$ by Lem. 4.2, so $\mathcal{E}$ misses at most c of the minimal FDs of Σ', which are the minimal FDs of the copies, and the declared key dependencies imply every FD of some copy. Otherwise, the restriction of $\mathcal{E}$ to every copy is enforceable for that copy by the proof of Lem. 4.2, so it misses a minimal FD of the copy, and the declared key dependencies imply the FD set of no copy. Attribute closures decide which case applies in polynomial time, so $\mathcal{A}$ decides 0-loss BCNF. □

Proof of Theorem 4.3. We reduce the maximum independent set problem, which is NP-complete [22], to the decision whether $\lambda(R, \Sigma) \leq k$. Let $G = (V, E)$ be a graph with n vertices and m edges, and let α be the size of a maximum independent set of G. Let R consist of attributes P_v and Q_v for every vertex v, and of attributes G_e, C_e, D^1_e, D^2_e and D^3_e for every edge e. Let Σ consist of the *vertex FD* $a_v = P_v \to Q_v$ for every vertex v and, for every edge $e = uv$, of the *edge FD* $b_e = Q_u Q_v G_e \to C_e$, the *blockers* $s^u_e = P_u G_e \to C_e$ and $s^v_e = P_v G_e \to C_e$, and the *victims* $h^i_e = C_e P_u P_v D^i_e \to G_e$ for $i = 1, 2, 3$. The schema (R, Σ) has $2n + 5m$ attributes and $n + 6m$ FDs, so it is computed in polynomial time. We show that $\Sigma = \Sigma_a$ and $\lambda(R, \Sigma) = 2m + n - \alpha$, so that G has an independent set of size α' iff $\lambda(R, \Sigma) \leq 2m + n - \alpha'$.

Atomic cover. The attributes P_v and D^i_e occur on no right-hand side of Σ, and Q_v occurs on the right-hand side of a_v only, so a_v is the only minimal FD with right-hand side Q_v. The attributes C_e and G_e occur only in the six FDs of the edge e, each of which has one of them on its left-hand side and the other on its right-hand side, so a set $X \subseteq R$ determines neither unless it contains one of them. Hence, X determines G_e iff $G_e \in X$ or X contains C_e, P_u, P_v and some D^i_e, and X determines C_e iff $C_e \in X$ or X contains G_e and P_u, or G_e and P_v, or G_e and one of P_u and Q_u and one of P_v and Q_v. Consequently, the victims are the minimal FDs with right-hand side G_e, the edge FD and the blockers are those with right-hand side C_e, and $\Sigma = \Sigma_a$. In particular, $P_u P_v G_e \to C_e$ is implied by Σ but not minimal, so it does not count towards the loss.

Lower bound. Let $\mathcal{E}$ be enforceable for (R, Σ), let Ψ be the set of its uncritical minimal FDs, which is a cover of $\mathcal{E}$ by Thm. 3.18, and let S be the set of vertices v with $a_v \in \mathcal{E}$. We first show that no victim h^i_e belongs to $\mathcal{E}$ if $\mathcal{E}$ contains one of s^u_e, s^v_e and $P_u P_v G_e \to C_e$. Suppose that $h^i_e \in \mathcal{E} = \Psi^+$. Then Ψ contains an FD $Z \to G_e$ with $Z \subseteq (C_e P_u P_v D^i_e)^+_{\mathcal{E}}$, and $Z \to G_e \in \Sigma^+$ shows $Z \supseteq C_e P_u P_v D^j_e$ for some j, where $j = i$ since $(C_e P_u P_v D^i_e)^+_\Sigma = C_e P_u Q_u P_v Q_v D^i_e G_e$. Each of the three FDs has its attributes in $Z G_e$ and a left-hand side whose closure under Σ, and hence under $\mathcal{E}$, does not contain D^i_e. Hence, if one of them belongs to $\mathcal{E}$, then $(Z G_e, \mathcal{E}[Z G_e])$ is not in BCNF, which contradicts the uncriticality of $Z \to G_e$. Now consider an edge $e = uv$ and its six FDs. If $s^u_e \in \mathcal{E}$ or $s^v_e \in \mathcal{E}$, then the three victims are not in $\mathcal{E}$. Otherwise, s^u_e and s^v_e are not in $\mathcal{E}$, and if $u, v \in S$, then either $b_e \notin \mathcal{E}$ or $P_u P_v G_e \to C_e \in \mathcal{E}$, which again excludes the

victims. In every case, $\mathcal{E}$ misses at least two of the six FDs, and at least three if $u, v \in S$. Together with the $n - |S|$ vertex FDs outside $\mathcal{E}$, this gives $|\Sigma_a - \mathcal{E}| \geq n - |S| + 2m + e(S)$ where $e(S)$ is the number of edges with both vertices in S. Removing one vertex of every such edge from S leaves an independent set of size at least $|S| - e(S)$, so $|\Sigma_a - \mathcal{E}| \geq 2m + n - \alpha$.

Upper bound. Let I be an independent set of G, let Δ consist of the blockers of all edges and of the vertex FDs a_v with $v \notin I$, and let $\mathcal{E} = (\Sigma - \Delta)^+$. Then $\mathcal{E}$ contains no element of Δ, since Q_v occurs on no right-hand side of $\Sigma - \Delta$ for $v \notin I$, and $(P_u G_e)^+_{\mathcal{E}} \subseteq P_u Q_u G_e$ because b_e requires Q_v as well. Every element of $\Sigma - \Delta$ is uncritical for $\mathcal{E}$. This is clear for the vertex FDs, whose schemata have two attributes, and for the edge FD b_e, since the only non-trivial FDs of $\mathcal{E}$ on $Q_u Q_v G_e C_e$ have a left-hand side that contains $Q_u Q_v G_e$. For a victim h^i_e, the non-trivial FDs of $\mathcal{E}$ on its schema $C_e P_u P_v D^i_e G_e$ have a left-hand side that contains $C_e P_u P_v D^i_e$: the attributes P_u, P_v and D^i_e occur on no right-hand side of $\Sigma - \Delta$, so G_e is determined only by sets that contain $C_e P_u P_v D^i_e$, and C_e only through b_e, which requires Q_u and Q_v and thus a_u and a_v in $\mathcal{E}$, while I contains at most one of u and v. By Thm. 3.18, $\mathcal{E}$ is enforceable, and its loss is $|\Delta| = 2m + n - |I|$. Hence, $\lambda(R, \Sigma) = 2m + n - \alpha$. □

Proof of the last statement of Corollary 4.4. Minimum vertex cover is APX-complete on graphs of maximum degree three [3], so there is a constant $\epsilon_0 > 0$ such that, unless P equals NP, no polynomial-time algorithm computes for every such graph a vertex cover of size at most $(1 + \epsilon_0)\tau$, where τ is the size of a minimum vertex cover. Let $\epsilon = \epsilon_0/7$, and suppose that a polynomial-time algorithm $\mathcal{A}$ returns for every schema (R, Σ) with $\Sigma = \Sigma_a$ a BCNF design with loss at most $(1 + \epsilon) \cdot \lambda(R, \Sigma)$. Given a graph $G = (V, E)$ of maximum degree three with n vertices and m edges, we construct the schema (R, Σ) of the proof of Thm. 4.3 in polynomial time. Then $\Sigma = \Sigma_a$ and $\lambda(R, \Sigma) = 2m + n - \alpha = 2m + \tau$, since the vertex covers of G are the complements of its independent sets. Let $\mathcal{E}$ be the enforced FDs of the design that $\mathcal{A}$ returns, and let S be the set of vertices v with $a_v \in \mathcal{E}$, which attribute closures under the declared key dependencies determine in polynomial time. Removing one vertex of every edge with both vertices in S leaves an independent set I with $|I| \geq |S| - e(S)$, so $V - I$ is a vertex cover of size at most $n - |S| + e(S)$. Since $\mathcal{E}$ is enforceable, the lower bound in the proof of Thm. 4.3 gives $|\Sigma_a - \mathcal{E}| \geq n - |S| + 2m + e(S)$, so $|V - I| \leq |\Sigma_a - \mathcal{E}| - 2m$. Every vertex covers at most three edges, so $m \leq 3\tau$, and $|V - I| \leq (1+\epsilon)(2m+\tau) - 2m = \tau + \epsilon(2m+\tau) \leq (1+7\epsilon)\tau = (1+\epsilon_0)\tau$. Hence, $\mathcal{A}$ yields a polynomial-time algorithm that computes a vertex cover of size at most $(1 + \epsilon_0)\tau$ for every graph of maximum degree three, so P equals NP. □

Proof of Theorem A.8. We reduce from the unsatisfiability of a formula φ in conjunctive normal form with variables $x_1, \ldots, x_n$, $n \geq 1$, and non-empty clauses $c_1, \ldots, c_m$. Let R consist of the *literal attributes* P_i and Q_i for x_i and $\neg x_i$, collected in L, the *clause attributes* $C_1, \ldots, C_m$, collected in C, and the attributes T and Z, and let $X = LZ$. Let Σ consist of $f = X \to T$, of $P_i Q_i T \to X$ for every i, of $T\ell \to C_j$ for every literal attribute ℓ of a literal in c_j, of $C \to Z$, and of $T \to Z$, and let $\Delta = \{T \to Z\}$.

Atomic cover. Without T, only $C \to Z$ and f apply, and the literal attributes are determined only by the FDs $P_i Q_i T \to X$. A case distinction over the right-hand sides then shows that Σ_a consists of $X \to T$, $LC \to T$, $T \to Z$, $C \to Z$, the FDs $T\ell \to C_j$ above, $P_i Q_i T \to C_j$ for every clause c_j without a literal of x_i, $X \to C_j$ for every j, and $P_i Q_i T \to B$ for every i and $B \in L - P_i Q_i$. Hence, Σ_a has $O(n^2 + nm + |\varphi|)$ elements and is computed in polynomial time, and we take it as the input. In particular, every minimal FD of Σ with right-hand side T has a left-hand side that contains X or LC.

Satisfiable formulas. Let $\mathcal{E}$ be the enforced FDs of a BCNF design with $\Sigma_a - \mathcal{E} = \Delta$, and let $S = XT$ be the schema of f. Let H be the set of literal attributes of the true literals of a satisfying assignment. Then $\mathcal{E} \supseteq \Sigma_a - \Delta$ implies $TH \to Z \in \mathcal{E}[S]$, while $(TH)^+_\Sigma = THCZ$ does not contain S, so $(S, \mathcal{E}[S])$ is not in BCNF. Since $\mathcal{E} \subseteq \Sigma^+$ contains f and $LC \to T$, these two FDs are the only minimal FDs of $\mathcal{E}$ with right-hand side T, and f is critical for $\mathcal{E}$. The FD $LC \to T$ is critical for $\mathcal{E}$ as well, since $T\ell \to C_1 \in \mathcal{E}[LCT]$ for a literal attribute ℓ of c_1, while $(T\ell)^+_\Sigma$ contains no literal attribute other than ℓ. Hence, no uncritical minimal FD of $\mathcal{E}$ has right-hand side T, the uncritical minimal FDs of $\mathcal{E}$ do not imply f, and $\mathcal{E}$ is not enforceable by Thm. 3.18, a contradiction. So no BCNF design, plain or not, leaves exactly Δ unenforced.

Unsatisfiable formulas. Let Ψ' consist of f, of the FDs $P_i Q_i T \to B$ in Σ_a, of the FDs $T\ell \to C_j$, and of $C \to Z$, and let $\mathcal{E} = \Psi'^+$. Then Ψ' implies every element of $\Sigma_a - \Delta$, since $LC \to T$ follows from $C \to Z$ and f, and the FDs $P_i Q_i T \to C_j$ and $X \to C_j$ follow through T and the FDs $T\ell \to C_j$, while $T^+_{\mathcal{E}} = T$, so $\mathcal{E} \cap \Sigma_a = \Sigma_a - \Delta$. Every element of Ψ' is uncritical for $\mathcal{E}$. On S, a subset without T determines no further attribute of S unless it contains X, and a subset Y with $T \in Y$ is a key of S if it contains some $P_i Q_i$, and otherwise determines within S at most Z, which it determines iff its literal attributes, which are consistent, satisfy every clause, which is impossible. On $P_i Q_i TB$, every subset without $P_i Q_i T$ determines no further attribute of the schema, since the literal attributes are determined only through some $P_k Q_k T$. On $T\ell C_j$ and on CZ, only the declared left-hand sides and their supersets determine further attributes of the schema. Hence, Ψ' is an uncritical set, and by Cor. 3.19, $\mathcal{E}$ is enforced by a BCNF design, which is plain and leaves exactly Δ unenforced. □

Proof of Theorem A.9. For the first statement, a nondeterministic algorithm guesses a closed FD set $\mathcal{E}$ by the table of the closures $X^+_{\mathcal{E}}$ for all $X \subseteq R$, of size $2^{|R|} \cdot |R|$. In time polynomial in $2^{|R|}$ and the size of the input, it verifies that the table is a closure operator, that $X^+_{\mathcal{E}} \subseteq X^+_\Sigma$ for every X, so that $\mathcal{E} \subseteq \Sigma^+$, computes Σ_a and $w(\Sigma_a - \mathcal{E})$, and tests the condition of Thm. 3.18 by computing the minimal FDs of $\mathcal{E}$, testing every schema XA for BCNF on its subsets, and comparing closures under the uncritical ones. For the second statement, some plain BCNF design has loss at most k iff there are a set $\Delta \subseteq \Sigma_a$ with $w(\Delta) \leq k$ and a set U of FDs such that, for $\Psi = \Sigma_a - \Delta$ and $\mathcal{E} = \Psi^+$, (i) $\mathcal{E} \cap \Delta = \emptyset$, (ii) every element of U is a minimal FD of $\mathcal{E}$, (iii) U implies Ψ, and (iv) every element $X \to A$ of U is uncritical for $\mathcal{E}$. Indeed, (ii) and (iii) give $U^+ = \mathcal{E}$, so (iv) and Thm. 3.18 make $\mathcal{E}$ enforceable, and (i) gives $\Sigma_a \cap \mathcal{E} = \Psi$, so the design has loss $w(\Delta)$ and is plain. Conversely, for a plain design with $\Sigma_a - \mathcal{E} = \Delta$, the uncritical minimal FDs of $\mathcal{E}$ imply every $g \in \Psi$, and the closure computation for g applies at most $|R|$ of them, so some such U has

at most $|\Psi| \cdot |R|$ elements. Conditions (i) to (iii) take polynomial time with closures under Ψ, and (iv) states that for every $Y \subseteq XA$ the set $Y^+_\Psi \cap XA$ is Y or XA, a coNP condition. Hence, the problem is in Σ^p_2. □

Proof of Theorem 4.5. Soundness is losslessness of $\mathcal{D}$ for the cover $\{K \to S \mid S \in \mathcal{D}, K \in \kappa(S)\}$ of $\mathcal{E}$, which the chase decides in polynomial time [1]. A witness against local enforcement is a schema $S \in \mathcal{D}$ and an FD $X \to A$ with $XA \subseteq S$ such that $X \to A \in \mathcal{E}$, which the attribute closure verifies in linear time, and $K \not\subseteq X$ for every $K \in \kappa(S)$, so local enforcement is in coNP. For the hardness, let φ be a propositional formula in conjunctive normal form with variables $x_1, \ldots, x_n$ and clauses $C_1, \ldots, C_m$, and let R consist of attributes P_i and Q_i for the literals x_i and $\neg x_i$ of every variable, an attribute C_j for every clause, and an attribute Z. Let $\mathcal{D}$ consist of the schema $S_0 = P_1Q_1 \cdots P_nQ_nZ$ with $\kappa(S_0) = \{P_iQ_i \mid 1 \leq i \leq n\}$, of a schema LC_j with $\kappa(LC_j) = \{L\}$ for every clause C_j and every attribute L of a literal in C_j, and of the schema $S_1 = C_1 \cdots C_mZ$ with $\kappa(S_1) = \{C_1 \cdots C_m\}$, and let Σ be the set of these key dependencies. Then $(\mathcal{D}, \kappa)$ is a key design of (R, Σ) with $\mathcal{E} = \Sigma^+$, its size is polynomial in the size of φ, and it is sound since S_0 contains the key P_1Q_1 of (R, Σ). No attribute C_j or Z occurs in a left-hand side of Σ other than $C_1 \cdots C_m$, so the only non-trivial FDs of $\mathcal{E}$ on a schema LC_j or on S_1 are implied by its declared key, and these schemata are locally enforced. Let $X \subseteq S_0$ contain no declared key of S_0, that is, at most one of P_i and Q_i for every i, so that the literals of X are consistent. The closure of X under Σ consists of X, of the attributes C_j of the clauses that contain a literal of X, and of Z iff every clause contains a literal of X. Hence, $\mathcal{E}$ contains a non-trivial FD on S_0 with left-hand side X iff $Z \notin X$ and the literals of X satisfy φ. Consequently, $(\mathcal{D}, \kappa)$ is locally enforced iff φ is unsatisfiable. □

Proof of Theorem 5.2. For the first statement, let $R = \{A, B, C, D, E\}$ and $\Sigma = \{D \to C, BE \to C, BD \to A, ABC \to D\}$. The minimal keys of (R, Σ) are ABE and BDE, and Σ_a consists of $BD \to A$, $D \to C$, $BE \to C$, $ABC \to D$ and $ABE \to D$. The design $\mathcal{D} = \{ABCD, BCE, BDE\}$ with $\kappa(ABCD) = \{ABC, BD\}$, $\kappa(BCE) = \{BE\}$ and $\kappa(BDE) = \{BDE\}$ enforces $\mathcal{E} = \{ABC \to D, BD \to AC, BE \to C\}^+$. It is sound since BDE is a key of $(R, \mathcal{E})$, it is locally enforced since every non-trivial FD of $\mathcal{E}$ on a schema has a declared key as left-hand side, and its loss is one since $\mathcal{E}$ contains $ABE \to D$ but not $D \to C$. The loss is minimum because Σ^+ is not enforceable: $ABC \to D$ and $ABE \to D$ are critical for Σ^+, since $D \to C$ is a non-key FD of $(ABCD, \Sigma[ABCD])$ and $BD \to A$ is a non-key FD of $(ABDE, \Sigma[ABDE])$, and the uncritical minimal FDs $BD \to A$, $D \to C$ and $BE \to C$ do not imply $ABC \to D$. A plain design with loss one enforces $\Psi = \Sigma_a - \{\varphi\}$ for some minimal FD φ with $\varphi \notin \Psi^+$ and Ψ^+ enforceable. For $\varphi = ABE \to D$, the FDs $BE \to C$ and $ABC \to D$ imply φ. For $\varphi = D \to C$, the FD $ABC \to D$ is critical for Ψ^+ since $BD \to A$ is a non-key FD of $(ABCD, \Psi^+[ABCD])$, and the uncritical minimal FDs $BD \to A$ and $BE \to C$ of Ψ^+ do not imply it. For $\varphi = ABC \to D$, the FD $ABE \to D$ is critical for Ψ^+ since $BD \to A$ is a non-key FD of $(ABDE, \Psi^+[ABDE])$, and no other minimal FD of Ψ^+ has D as right-hand side. For $\varphi = BE \to C$, both $ABC \to D$ and $ABE \to D$ are critical for Ψ^+ by the same two non-key FDs, and no other minimal FD of Ψ^+ has D as right-hand side. For $\varphi = BD \to A$, the FD $ABC \to D$ is critical for Ψ^+ since $D \to C$ is a non-key FD of $(ABCD, \Psi^+[ABCD])$, and the uncritical minimal FDs $ABE \to D$, $D \to C$ and $BE \to C$ of Ψ^+ do not imply it. Hence, every plain design has a loss of at least two.

For the second statement, let $R = \{A, B, C, D, E\}$ and $\Sigma = \{B \to E, CE \to A, CE \to B, ADE \to B, ABD \to C\}$. The minimal keys of (R, Σ) are ABD, ADE, BCD and CDE, and Σ_a consists of $B \to E$, $BC \to A$, $CE \to A$, $CE \to B$, $ADE \to B$, $ABD \to C$ and $ADE \to C$. The design $\mathcal{D} = \{ACE, BCE, ABDE, ABCD\}$ with the single keys CE, CE, ADE and ABD enforces $\mathcal{E} = \{CE \to AB, ADE \to B, ABD \to C\}^+$, which contains $ADE \to C$ but neither $B \to E$ nor $BC \to A$. It is sound since ADE is a key of $(R, \mathcal{E})$ contained in $ABDE$, it is locally enforced since every non-trivial FD of $\mathcal{E}$ on a schema has the declared key as left-hand side, and its loss is two. The loss is minimum. Indeed, each of $BC \to A$, $CE \to A$, $ADE \to B$, $ABD \to C$ and $ADE \to C$ is implied by the other minimal FDs, so a design that leaves only one of them unenforced does not exist. A design that leaves only $B \to E$ unenforced enforces $BC \to A$ and $CE \to A$, which remain non-key FDs of every schema that contains $ABCD$ or $ACDE$, so every minimal FD with right-hand side C is critical and $ABD \to C$ cannot be enforced. A design that enforces $B \to E$, in particular one with loss zero or one that leaves only $CE \to B$ unenforced, has the non-key FD $B \to E$ on every schema that contains BE properly. Every minimal FD of its enforced FDs with right-hand side B has E on its left-hand side and is therefore critical, so by Thm. 3.18 the design enforces neither $CE \to B$ nor $ADE \to B$. Hence, $\lambda(R, \Sigma) = 2$. Now let $(\mathcal{D}', \kappa')$ be a key-complete BCNF design with enforced FDs $\mathcal{E}'$, and suppose its loss is at most two. We use that C and E determine nothing but themselves, that CE determines $ABCE$ only, and that B determines BE only. Suppose first that $B \to E \in \mathcal{E}'$. Then no schema of $\mathcal{D}'$ contains BE properly, since B is not a key of such a schema. The key dependencies of $\mathcal{D}'$ derive B from CE or from ADE only through a schema that contains BE properly, so $CE \to B$ and $ADE \to B$ are both unenforced. They derive C from ABD only through a schema that contains $ABCD$ or $ACDE$, on which $BC \to A$ or $CE \to A$ would violate local enforcement, so a third minimal FD is unenforced. Suppose next that $B \to E \notin \mathcal{E}'$ and $CE \to B \in \mathcal{E}'$. Then some schema $S^* \in \{BCE, ABCE\}$ with $CE \in \kappa'(S^*)$ belongs to $\mathcal{D}'$. Since BC is a minimal key of $(S^*, \Sigma[S^*])$, key-completeness gives $BC \in \kappa'(S^*)$ and $BC \to E \in \mathcal{E}'$. If $CE \to A \notin \mathcal{E}'$, then $BC \to A \notin \mathcal{E}'$ either, as $BC \to A$ and $CE \to B$ would imply $CE \to A$, and three minimal FDs are unenforced. If $CE \to A \in \mathcal{E}'$, then $BC \to A \in \mathcal{E}'$, and C is derived from ABD only through a schema that contains $ABCD$ or $ACDE$, which $BC \to A$ or $CE \to A$ prevent, so $ABD \to C$ and $ADE \to C$ are unenforced. Suppose finally that neither $B \to E$ nor $CE \to B$ is in $\mathcal{E}'$. Then $BC \to A$ and $CE \to A$ must be enforced, and C is again derived from ABD only through a schema that contains $ABCD$ or $ACDE$, so $ABD \to C$ is unenforced. In all cases the loss is at least three. □

Proof of Proposition A.7. Every plain BCNF design is a BCNF design, which gives the first inequality. The uncritical design of Thm. 3.22 enforces $\mathcal{E} = \mathcal{U}^+$, and $\mathcal{U} \subseteq \Sigma_a \cap \mathcal{E}$ gives $(\Sigma_a \cap \mathcal{E})^+ = \mathcal{E}$, so it is a plain BCNF design with loss $w(O)$, which gives the second inequality.

For the second statement, let $\mathcal{V} = \{C_1, \ldots, C_m\}$, let $\mathcal{Y}$ be the set of subsets of $\mathcal{V}$ with $\lfloor m/2 \rfloor$ elements, which is an antichain with $|\mathcal{Y}| = \binom{m}{\lfloor m/2 \rfloor} \geq 2$ whose elements are non-empty proper subsets of $\mathcal{V}$, let $R_m = \{A, B, E\} \cup \mathcal{V}$, and let $\Sigma_m = \{B \to A\} \cup \{BY \to E, AYE \to B \mid Y \in \mathcal{Y}\}$. No attribute of $\mathcal{V}$ occurs on a right-hand side, A occurs only on the right-hand side of $B \to A$, E only on those of the FDs $BY \to E$, and B only on those of the FDs $AYE \to B$. Hence, a set $X \subseteq R_m$ determines A iff it contains A or B, it determines E iff it contains E or some BY with $Y \in \mathcal{Y}$, and it determines B iff it contains B or some AYE with $Y \in \mathcal{Y}$. Since $\mathcal{Y}$ is an antichain, $\Sigma_a = \Sigma_m$, $B^+_{\Sigma_m} = AB$, $(BY)^+_{\Sigma_m} = (AYE)^+_{\Sigma_m} = ABYE$ for every $Y \in \mathcal{Y}$, and no subset of $\mathcal{V}$ determines A. The FD $B \to A$ is uncritical since its schema has two attributes, and so is $BY \to E$, since every non-trivial FD of Σ^+_m on BYE has a left-hand side that contains BY' for some $Y' \in \mathcal{Y}$ with $Y' \subseteq Y$, that is, $Y' = Y$. The FD $AYE \to B$ is critical, since $B \to A$ is a non-key FD of $(ABYE, \Sigma_m[ABYE])$, and the uncritical minimal FDs do not imply it, since the FDs $AY'E \to B$ are the only minimal FDs with right-hand side B. Hence, the obstructions of (R_m, Σ_m) are the $|\mathcal{Y}|$ FDs $AYE \to B$.

We first show that every enforceable $\mathcal{E}$ that contains $AYE \to B$ has it as an uncritical minimal FD. By Lem. 3.3, every minimal FD of $\mathcal{E}$ with right-hand side B has the form $X \to B$ with $AY'E \subseteq X$ for some $Y' \in \mathcal{Y}$, so $AYE \to B$ is a minimal FD of $\mathcal{E}$, as $\mathcal{Y}$ is an antichain. By Thm. 3.18, the uncritical minimal FDs of $\mathcal{E}$ imply it, so the closure of AYE under them applies an uncritical minimal FD $X \to B$ of $\mathcal{E}$ with $X \subseteq (AYE)^+_{\mathcal{E}} \subseteq ABYE$. Since $B \notin X$ and X contains some $AY'E$, we obtain $Y' \subseteq Y$, hence $Y' = Y$ and $X = AYE$.

Next, let $\mathcal{E}$ be enforceable. If $B \to A \in \mathcal{E}$, then $\mathcal{E}$ contains no FD $AYE \to B$, since $B \to A$ would be a non-key FD of $(ABYE, \mathcal{E}[ABYE])$ by $B^+_{\mathcal{E}} \subseteq B^+_{\Sigma_m} = AB$, which contradicts the uncriticality of $AYE \to B$. Hence, every BCNF design that enforces $B \to A$ has a loss of at least $|\mathcal{Y}| \geq 2$, so a design with loss one enforces every minimal FD of Σ_m except $B \to A$. For such a design and every $Y \in \mathcal{Y}$, the FD $AYE \to B$ is uncritical for $\mathcal{E}$, and $BY \to E$ is a non-trivial FD of $\mathcal{E}[ABYE]$, so BY is a key of $ABYE$ for $\mathcal{E}$ and $BY \to A \in \mathcal{E}$. This FD is a minimal FD of $\mathcal{E}$, since $B \to A \notin \mathcal{E}$, no subset of $\mathcal{V}$ determines A, and a set BY'' with $Y'' \subset Y$ contains no BY' with $Y' \in \mathcal{Y}$, so its closure under $\mathcal{E}$ does not contain A. Hence, the design enforces the $|\mathcal{Y}|$ weakened forms $BY \to A$ of $B \to A$. In particular, Σ^+_m is not enforceable and $\lambda(R_m, \Sigma_m) \geq 1$.

The design with the schemata $ABYE$ for $Y \in \mathcal{Y}$, each declaring the keys BY and AYE, and the schema $K = B\mathcal{V}$ declaring the key K has loss one. Its enforced FDs $\mathcal{E} = \{BY \to AE, AYE \to B \mid Y \in \mathcal{Y}\}^+$ contain every minimal FD of Σ_m except $B \to A$. A subset of $ABYE$ that contains neither BY nor AYE is closed under $\mathcal{E}$, since every other declared key contains an attribute of $\mathcal{V} - Y$, and the closure of a subset of K adds at most A and E, which lie outside K. Hence, the design is locally enforced, and it is sound since it is dependency-preserving for $\mathcal{E}$ and contains the key K of $(R_m, \mathcal{E})$ [13]. Thus, $\lambda(R_m, \Sigma_m) = 1$.

Finally, let $\mathcal{E}$ be the enforced FDs of a plain BCNF design with $B \to A \notin \mathcal{E}$. Then $\mathcal{E}$ is generated by minimal FDs of Σ_m other than $B \to A$, none of which has A as right-hand side, so a set X determines A under $\mathcal{E}$ only if $A \in X$. If $\mathcal{E}$ contained both $BY \to E$ and $AYE \to B$ for some $Y \in \mathcal{Y}$, then $AYE \to B$ would be uncritical for $\mathcal{E}$, so BY would be a key of $ABYE$ for $\mathcal{E}$ and determine A, a contradiction. Hence, the design leaves $B \to A$ and one of $BY \to E$ and $AYE \to B$ for every $Y \in \mathcal{Y}$ unenforced, and its loss is at least $|\mathcal{Y}| + 1$. Together with the case $B \to A \in \mathcal{E}$, every plain BCNF design has a loss of at least $|\mathcal{Y}|$, and $\lambda^{\mathrm{p}}(R_m, \Sigma_m) = |\mathcal{O}| = |\mathcal{Y}|$ by the first statement. □

Proof of Lemma 5.3. Let $\mathcal{U}_{\mathcal{E}}$ denote the uncritical minimal FDs of $\mathcal{E}$, which form a cover of $\mathcal{E}$ by Thm. 3.18. Since $\Sigma_a \subseteq \mathcal{E} \cup \Delta$, the set $\mathcal{U}_{\mathcal{E}} \cup \Delta$ is a cover of Σ, so it suffices to show that $\mathcal{U} \cup \mathcal{C}(\Delta) \cup \Delta$ implies every $Y \to B \in \mathcal{U}_{\mathcal{E}}$. Firstly, let $Y \to B$ be a minimal FD of Σ that is critical for Σ^+, so that some minimal FD of Σ violates YB. If some such FD $Z \to C$ belonged to $\mathcal{E}$, then $Z \to C \in \mathcal{E}[YB]$ where Z is not a key of YB for $\mathcal{E} \subseteq \Sigma^+$, so $(YB, \mathcal{E}[YB])$ would not be in BCNF, which contradicts $Y \to B \in \mathcal{U}_{\mathcal{E}}$. Hence, every minimal FD of Σ that violates YB belongs to Δ, that is, $Y \to B \in \mathcal{C}(\Delta)$. Secondly, let $Y \to B$ not be a minimal FD of Σ. Then $Y' \to B$ is a minimal FD of Σ for some $Y' \subset Y$, and $Y' \to B \notin \mathcal{E}$ since $Y \to B$ is minimal for $\mathcal{E}$. Hence, $Y' \to B \in \Delta$ implies $Y \to B$. Every remaining element of $\mathcal{U}_{\mathcal{E}}$ is a minimal FD of Σ that is uncritical for Σ^+, that is, an element of $\mathcal{U}$. For the second claim, suppose that Δ contains neither a critical minimal FD of Σ nor an FD that violates the schema of one. Then $\mathcal{C}(\Delta) = \emptyset$ and $\Delta \subseteq \mathcal{U}$, so $\mathcal{U}$ is a cover of Σ and Σ^+ is enforceable by Thm. 3.18. □

Proof of Proposition 5.4. Let $(\mathcal{D}, \kappa)$ be a BCNF design of (R, Σ) with enforced FDs $\mathcal{E}$ such that $\Delta' = \Sigma_a - \mathcal{E}$ contains Δ, and let $g_i = Y \to B$. By Lem. 5.3, g_i is implied by $\mathcal{U} \cup \mathcal{C}(\Delta') \cup \Delta'$, and the attribute closure of Y under this set applies only FDs whose left-hand sides are contained in Y^+_Σ. Since $\mathcal{U} \cup \mathcal{C}(\Delta) \cup \Delta$ does not imply g_i, some FD f applied in the closure is not in this set, and $f \in \mathcal{C}(\Delta') \cup \Delta'$. If $f \in \Delta'$, then f belongs to $\mathcal{Q}(g_i, \Delta)$ by the first clause of its definition. Otherwise, $f = X \to A$ is a critical minimal FD of Σ with $X \subseteq Y^+_\Sigma$ and $f \in \mathcal{C}(\Delta') - \mathcal{C}(\Delta)$, so every minimal FD of Σ that violates XA belongs to Δ', and some such FD v is not in Δ since $f \notin \mathcal{C}(\Delta)$. Hence, v belongs to $\mathcal{Q}(g_i, \Delta)$ by the second clause, and $v \in \Delta'$. In both cases Δ' meets $\mathcal{Q}(g_i, \Delta)$, which is impossible if the requirement set is empty. Since the requirement sets are pairwise disjoint and disjoint from Δ, the set Δ' contains Δ and one distinct element of every $\mathcal{Q}(g_i, \Delta)$, which gives the bound on $w(\Delta')$. □

Proof of Theorem 5.5. A returned pair $(\mathcal{D}, \kappa)$ is a key design since every minimal key of $(S, \mathcal{E}[S])$ is a key of S for Σ. It is sound and locally enforced by Thm. 3.18 applied to $\mathcal{E}$, and its loss is $w(\Delta)$ by line 7. For optimality, let $\mathcal{E}^*$ be the enforced FDs of a BCNF design with minimum loss, and let $\Delta^* = \Sigma_a - \mathcal{E}^*$. By Lem. 3.3, every minimal FD of $\mathcal{E}^*$ either belongs to $\Sigma_a - \Delta^*$ or is a weakening of some minimal FD $X \to A$ of Σ with $X \to A \notin \mathcal{E}^*$, that is, an element of $\mathcal{W}(\Delta^*)$. Hence, $\mathcal{E}^* = (\Psi^* \cup W^*)^+$ for $\Psi^* = \Sigma_a - \Delta^*$ and the set W^* of minimal FDs of $\mathcal{E}^*$ in $\mathcal{W}(\Delta^*)$, and $\mathcal{E}^* \cap \Delta^* = \emptyset$. By Lem. 5.3, Δ^* passes the test in line 3. Since $\mathcal{E}^*$ is enforceable, the combination of Δ^* and W^* causes a return, unless the algorithm returns earlier with a loss of at most $w(\Delta^*)$. The restriction to $W = \emptyset$ examines the sets Ψ^+ with $\Psi \subseteq \Sigma_a$ and $\Psi^+ \cap \Sigma_a = \Psi$ whose complement $\Sigma_a - \Psi$ passes line 3, and these include the enforced FDs of every plain design by Lem. 5.3. □

Proof of Lemma A.14. Let $\mathcal{U}_{\mathcal{E}}$ and $\mathcal{U}_{\mathcal{E}'}$ denote the uncritical minimal FDs of $\mathcal{E}$ and $\mathcal{E}'$. By Thm. 3.18, $\mathcal{U}_{\mathcal{E}'}$ implies g, so the computation of the closure of Y under $\mathcal{U}_{\mathcal{E}'}$ derives B. Since $\mathcal{U}_{\mathcal{E}}$ does not imply g, this computation applies some FD of $\mathcal{U}_{\mathcal{E}'} - \mathcal{U}_{\mathcal{E}}$, and we let $f = V \to C$ be the first one. The FDs applied before f belong to $\mathcal{U}_{\mathcal{E}} \subseteq \mathcal{E}$, so $V \subseteq Y^+_{\mathcal{E}}$. Firstly, let $f \notin \mathcal{E}$. By Lem. 3.3, some minimal FD $X \to C$ of Σ satisfies $X \subseteq V$. It belongs to Δ, since otherwise $X \to C \in \mathcal{E}$ and hence $f \in \mathcal{E}$, and $X \neq V$, since $f \in \mathcal{E}'$ and $\mathcal{E}' \cap \Delta = \emptyset$. Hence, f is the required FD. Secondly, let $f \in \mathcal{E}$. Then f is a minimal FD of $\mathcal{E}$, since some $V' \subset V$ with $V' \to C \in \mathcal{E} \subseteq \mathcal{E}'$ would contradict the minimality of f in $\mathcal{E}'$, and f is critical for $\mathcal{E}$ since $f \notin \mathcal{U}_{\mathcal{E}}$. So some non-trivial FD $Z \to D \in \mathcal{E}[VC]$ has a left-hand side Z that is not a key of VC for $\mathcal{E}$, while Z is a key of VC for $\mathcal{E}'$ since f is uncritical for $\mathcal{E}'$. Hence, $Z \to a \in \mathcal{E}' - \mathcal{E}$ for some $a \in VC$, and a minimal $Z' \subseteq Z$ with $Z' \to a \in \mathcal{E}'$ yields a minimal FD $Z' \to a$ of $\mathcal{E}'$ that is not in $\mathcal{E}$. As before, Lem. 3.3 provides some $X \to a \in \Delta$ with $X \subset Z'$, and $Z' \subseteq VC \subseteq Y^+_{\mathcal{E}}$ since $V \subseteq Y^+_{\mathcal{E}}$ and $f \in \mathcal{E}$. □

Proof of Lemma 5.6. Let $\mathcal{E}$ be the enforced FDs of a BCNF design and $\Delta' = \Sigma_a - \mathcal{E}$. For statement 1, if $\Delta \subseteq \Delta'$, then Δ' meets $\mathcal{Q}(g, \Delta)$ by Prop. 5.4. For statement 2, if $f \in \Delta'$, then $D \not\subseteq \mathcal{E}$, since $\mathcal{E}$ is closed under implication and $f \notin \mathcal{E}$, so D meets Δ'. For statement 3, $\Delta' \neq \Delta$ by assumption, so $\Delta \not\subseteq \Delta'$ or Δ' contains an element of $\Sigma_a - \Delta$. □

Proof of Theorem 5.7. Every cut in $\mathcal{K}$ is valid. The initial cuts are of the first kind of Lem. 5.6 with $\Delta = \emptyset$, since the obstructions are the minimal FDs not implied by $\mathcal{U}$ and $C(\emptyset) = \emptyset$. The cuts added after line 5 are of the second kind. The cuts added in line 10 are of the first kindfor the subsets $\Delta \cap \mathcal{N}(g)$ of Δ, as shown after Lem. 5.6, and, since line 7 failed and every BCNF design that leaves exactly Δ unenforced enforces an enforceable $\mathcal{E} \supseteq \Psi$ with $\mathcal{E} \cap \Delta = \emptyset$, of the third kind. Hence, the set $\Sigma_a - \mathcal{E}^*$ of a BCNF design with minimum loss satisfies $\mathcal{K}$ in every iteration, and $w(\Delta) \leq \lambda_w(R, \Sigma)$ in line 3. Every iteration that does not return adds a cut that Δ violates: a cut $(\{f\}, D_f)$ with $f \in \Delta$ and $D_f \cap \Delta = \emptyset$, or the cut $(\Delta, \Sigma_a - \Delta)$. Hence, no set Δ is computed twice, and the algorithm terminates after at most $2^{|\Sigma_a|}$ iterations, since the master problem remains feasible. When line 7 succeeds, $\mathcal{E}$ is enforceable with $\Sigma_a \cap \mathcal{E} = \Psi$, so Cor. 3.19 and Thm. 3.18 yield a BCNF design with enforced FDs $\mathcal{E}$ and loss $w(\Delta) \leq \lambda_w(R, \Sigma)$, which is therefore minimum. The search in line 7 is complete by Lem. A.14, as in the proof of Thm. 5.5. For the restriction to $\mathcal{E} = \Psi^+$, the cuts of the first two kinds are valid for plain designs as for all BCNF designs, and the cuts of the third kind are valid for plain designs, since a plain BCNF design that leaves exactly Δ unenforced enforces Ψ^+. The same argument then shows termination, the lower bounds and the minimality among plain designs. □

Proof of Proposition A.3. For the first statement, let $R = \{A, B, C, D, E\}$ and $\Sigma = \{B \to A, BC \to E, BD \to E, ACE \to B, ADE \to B\}$, whose minimal FDs are the elements of Σ. The design $\{ABCE, ABDE, BCD\}$ with $\kappa(ABCE) = \{ACE, BC\}$, $\kappa(ABDE) = \{ADE, BD\}$ and $\kappa(BCD) = \{BCD\}$ enforces $\mathcal{E} = \{ACE \to B, ADE \to B, BC \to AE, BD \to AE\}^+$. It is sound since BCD is a key of $(R, \mathcal{E})$, it is locally enforced since every non-trivial FD of $\mathcal{E}$ on a schema has a declared key as left-hand side, and its loss is one since $B \to A$ is the only minimal FD outside $\mathcal{E}$. Every BCNF design with loss one leaves $B \to A$ unenforced. Indeed, if one of the other four minimal FDs is the only unenforced one, then $B \to A$ is a non-key FD of $ABCE$ and of $ABDE$, so every FD with right-hand side B is critical and $ACE \to B$ or $ADE \to B$ cannot be enforced. With $B \to A$ unenforced, $ACE \to B$ and $ADE \to B$ are the only FDs with right-hand side B, so both must be uncritical, which requires that BC is a key of $ABCE$ and that BD is a key of $ABDE$. Hence, $BC \to A$ and $BD \to A$ are enforced, and both are minimal FDs of the enforced set since neither C nor D determines A.

For the second statement, let $R = \{A, B, C, D, E\}$ and $\Sigma = \{A \to D, DE \to A, AE \to B, BCE \to D\}$, whose minimal FDs are the elements of Σ together with $DE \to B$ and $BCE \to A$. The design $\{ABCE, ABD, ADE, BDE\}$ with the single keys BCE, AB, DE and DE enforces $\mathcal{E} = \{BCE \to A, AB \to D, DE \to AB\}^+$, which contains $BCE \to D$ but neither $A \to D$ nor $AE \to B$. It is sound since BCE is a key of $(R, \mathcal{E})$, it is locally enforced, and its loss is two. An exhaustive enumeration of the closed FD sets between Σ^+ and the trivial FD set shows that every enforceable FD set with loss two leaves exactly $A \to D$ and $AE \to B$ unenforced. Such an FD set contains $BCE \to D$, which is critical since $DE \to B$ is a non-key FD of $BCDE$, so $BCE \to D$ is derived from $BCE \to A$ and some weakened form $X' \to D$ of $A \to D$ with $A \in X' \subseteq ABCE$. If $X' \subseteq AE$, then $X' \to D$ and $DE \to B$ imply the unenforced FD $AE \to B$. Hence, X' contains B or C, and no minimal FD of Σ has a left-hand side that contains AB or AC. □

Proof of Proposition A.4. Let $R = \{A, B, C, D, E, F\}$ and $\Sigma = \{DF \to C, ACE \to D, AE \to B, AF \to D, BDE \to A, DE \to F\}$. The atomic cover Σ_a consists of the elements of Σ and the minimal FDs $DE \to C$, $AF \to C$ and $ACE \to F$. Let $\Delta = \{AE \to B, AF \to C, AF \to D\}$, let $\Psi = \Sigma_a - \Delta$, and let w assign weight one to the elements of Δ and weight four to the other six minimal FDs. A closure computation shows that the elements of Ψ are the minimal FDs of Ψ^+, that only $ACE \to D$ among them is critical for Ψ^+, since $DE \to C$ violates $ACDE$, and that $ACE^+_\Psi = ACDEF$.

We first show that an enforceable FD set $\mathcal{E} \subseteq \Sigma^+$ contains neither both $AE \to B$ and $BDE \to A$, nor all of $DE \to C$, $AF \to C$, $ACE \to D$ and $ACE \to F$. Since $\mathcal{E} \subseteq \Sigma^+$, the left-hand side of every non-trivial FD in $\mathcal{E}$ with right-hand side A contains BDE, with right-hand side D it contains ACE or AF, and with right-hand side F it contains DE or ACE. If $AE \to B$ and $BDE \to A$ are in $\mathcal{E}$, then $BDE \to A$ is the only minimal FD of $\mathcal{E}$ with right-hand side A, and it is critical for $\mathcal{E}$ since $AE \to B$ violates $ABDE$. Hence, the uncritical minimal FDs of $\mathcal{E}$ do not imply $BDE \to A$, and $\mathcal{E}$ is not enforceable by Thm. 3.18. If $DE \to C$, $AF \to C$, $ACE \to D$ and $ACE \to F$ are in $\mathcal{E}$, then $ACE \to D$ and $ACE \to F$ are minimal FDs of $\mathcal{E}$ that are critical for $\mathcal{E}$, since $DE \to C$ violates $ACDE$ and $AF \to C$ violates $ACEF$. Every other minimal FD of $\mathcal{E}$ with right-hand side D has F in its left-hand side, and vice versa. Hence, the uncritical minimal FDs of $\mathcal{E}$ do not imply $ACE \to D$, and $\mathcal{E}$ is not enforceable.

Next, the design $\mathcal{D} = \{ABDE, ABCE, CDE, ABCF, CDF, ABDF, ACEF, DEF\}$ with $\kappa(ABDE) = \{ADE, BDE\}$, $\kappa(ABCE) = \{ACE\}$, $\kappa(ABCF) = \kappa(ABDF) = \{ABF\}$ and the single keys DE, DF, ACE and DE of CDE, CDF, $ACEF$ and DEF enforces $\mathcal{E} = (\Psi \cup \{ACE \to$

$B, ADE \to B, ABF \to C, ABF \to D\})^+$. It is sound since ACE is a key of $(R, \mathcal{E})$ contained in $ACEF$ and $\mathcal{D}$ is dependency-preserving for $\mathcal{E}$, and it is locally enforced since every non-trivial FD of $\mathcal{E}$ on a schema has a declared key in its left-hand side. It leaves exactly Δ unenforced, since AE and AF are closed for $\mathcal{E}$, so its loss is $w(\Delta) = 3$. Now let $\mathcal{E}'$ be the enforced FDs of a BCNF design of (R, Σ) with loss less than four, so that its unenforced FDs form a subset Δ' of Δ. If $AE \to B \in \mathcal{E}'$, then $\mathcal{E}'$ contains $AE \to B$ and $BDE \to A$. If $AE \to B \notin \mathcal{E}'$ and $AF \to D \in \mathcal{E}'$, then $AF \to C \in \mathcal{E}'$ as well, so $\mathcal{E}'$ contains $DE \to C$, $AF \to C$, $ACE \to D$ and $ACE \to F$. If $AE \to B$ and $AF \to D$ are not in $\mathcal{E}'$ but $AF \to C$ is, then $\mathcal{E}'$ contains the same four FDs. In each case $\mathcal{E}'$ is not enforceable, so $\Delta' = \Delta$. Hence, $\lambda_w(R, \Sigma) = 3$, and every BCNF design with this loss leaves exactly Δ unenforced. Under unit weights, let Δ' be the set of unenforced FDs of a BCNF design with enforced FDs $\mathcal{E}'$, and suppose that $|\Delta'| \le 2$. Then Δ' contains $AE \to B$ or $BDE \to A$, and the other element of Δ', if any, is none of $DE \to C$, $AF \to C$, $ACE \to D$ and $ACE \to F$, since each of these is implied by the elements of $\Sigma_a - \Delta'$. Hence, $\mathcal{E}'$ contains these four FDs and is not enforceable, so $\lambda(R, \Sigma) = 3 = |\Delta|$.

Finally, let $\mathcal{E}'$ be the enforced FDs of a BCNF design of (R, Σ) that leaves exactly Δ unenforced. Then $ACE \to D$ is a minimal FD of $\mathcal{E}'$ that is critical for $\mathcal{E}'$, since $DE \to C \in \mathcal{E}'$. By Thm. 3.18, the uncritical minimal FDs of $\mathcal{E}'$ imply $ACE \to D$, so D is derived from ACE through some uncritical minimal FD $X \to D$ of $\mathcal{E}'$ with $X \ne ACE$. Then X does not contain ACE and is a proper superset of AF, since $X \to D$ is minimal and $AF \to D \notin \mathcal{E}'$. If $C \in X$, then $DF \to C$ violates XD, and if $E \in X$, then $DE \to F$ violates XD, since neither DF nor DE determines A for Σ. Hence, $X = ABF$, and $ABF \to D$ is not local since $B \notin ACE^+_\Psi$. □

Proof of Proposition A.5. Let $R = \{A, B, C, D, E\}$ and $\Sigma = \{A \to B, A \to D, D \to A, B \to E, E \to B, BC \to A\}$. The atomic cover Σ_a consists of $A \to B$, $A \to D$, $A \to E$, $D \to A$, $D \to B$, $D \to E$, $B \to E$, $E \to B$, $BC \to A$, $BC \to D$, $CE \to A$ and $CE \to D$. The minimal FDs with two attributes are uncritical. The minimal FD $BC \to A$ is critical since $A \to B$ violates ABC, and $A \to B$ is the only minimal FD that violates ABC. Likewise, $BC \to D$, $CE \to A$ and $CE \to D$ are critical, and their schemata are violated by $D \to B$, $A \to E$ and $D \to E$ only. Let G consist of these four critical minimal FDs and the four FDs that violate their schemata. The uncritical minimal FDs $\mathcal{U} = \Sigma_a - \{BC \to A, BC \to D, CE \to A, CE \to D\}$ do not imply $BC \to A$ since $BC^+_{\mathcal{U}} = BCE$, so Σ^+ is not enforceable by Thm. 3.18.

Let $\mathcal{D} = \{ABD, ADE, BE, BCD, CDE\}$ with $\kappa(ABD) = \{A, BD\}$, $\kappa(ADE) = \{A, DE\}$, $\kappa(BE) = \{B, E\}$, $\kappa(BCD) = \{BC\}$ and $\kappa(CDE) = \{CE\}$, and let $\mathcal{E}$ be the closure of the declared key dependencies. Then $D^+_{\mathcal{E}} = D$, so $D \to A$, $D \to B$ and $D \to E$ are not in $\mathcal{E}$, while every other minimal FD of Σ is, as $BC \to A$ follows from $BC \to D$ and $BD \to A$, and $CE \to A$ follows from $CE \to D$ and $DE \to A$. The decomposition is dependency-preserving for $\mathcal{E}$ and contains the key BC of $(R, \mathcal{E})$, so it is lossless for $\mathcal{E}$ and the design is sound. It is locally enforced since $\mathcal{E}[ABD] = \{A \to BD, BD \to A\}^+$, $\mathcal{E}[ADE] = \{A \to DE, DE \to A\}^+$, $\mathcal{E}[BE] = \{B \to E, E \to B\}^+$, $\mathcal{E}[BCD] = \{BC \to D\}^+$ and $\mathcal{E}[CDE] = \{CE \to D\}^+$, as C and D determine nothing and B and E determine only each other. Hence, $\lambda(R, \Sigma) \le 3$.

Let $\mathcal{E}'$ be enforceable for (R, Σ) with $\Delta = \Sigma_a - \mathcal{E}'$. Exchanging A with D or B with E maps Σ to a cover of itself, and the two exchanges act transitively on the critical minimal FDs and on the FDs that violate their schemata. By Lem. 5.3, Δ contains an element of G, so it suffices to consider $A \to B \in \Delta$ and $BC \to A \in \Delta$. If $A \to B \in \Delta$, then $\mathcal{E}'$ contains neither both $A \to D$ and $D \to B$ nor both $A \to E$ and $E \to B$, since $\mathcal{E}'$ is closed under implication and $A \to B \notin \mathcal{E}'$. If $BC \to A \in \Delta$, then $\mathcal{E}'$ contains neither both $BC \to D$ and $D \to A$ nor both $B \to E$ and $CE \to A$. In both cases $|\Delta| \ge 3$, so $\lambda(R, \Sigma) = 3$. Finally, let $|\Delta| = 3$ and $\Delta \subseteq G$. In the first case, $\Delta = \{A \to B, D \to B, A \to E\}$ since $A \to D$ and $E \to B$ are not in G, but $D \to E$ and $E \to B$ belong to $\mathcal{E}'$ and imply $D \to B$. In the second case, $\Delta = \{BC \to A, BC \to D, CE \to A\}$ since $D \to A$ and $B \to E$ are not in G, but $CE \to D$ and $D \to A$ belong to $\mathcal{E}'$ and imply $CE \to A$. Both cases contradict $\Delta = \Sigma_a - \mathcal{E}'$, so every design with loss three leaves an element of $\Sigma_a - G$ unenforced, and every element of $\Sigma_a - G$ is uncritical and violates the schema of no critical minimal FD. □

Proof of Proposition 5.8. The uncritical minimal FDs of the enforceable set $\mathcal{E}$ form a cover of $\mathcal{E}$ by Thm. 3.18, and they form an uncritical set, since each of them is uncritical for $\mathcal{E}$. Every set Ψ that the algorithm maintains is therefore uncritical: the initial set by the previous sentence, every set of a climb by the tests in lines 5 and 9, and every kick by line 12. Line 5 adds an FD $g \in \Sigma_a - \Psi^+$, which enlarges Ψ^+ and enforces g, so $w(\Sigma_a - \Psi^+)$ decreases by at least $w(g)$, and line 9 changes Ψ only if $w(\Sigma_a - \Psi^+)$ does not increase, and at most s times in a row without decreasing it. Since the subsets of Σ^+ are finite in number, every climb terminates. Line 12 replaces Ψ^* only by a set of at most its loss, so $w(\Sigma_a - \Psi^{*+})$ never increases from $w(\Sigma_a - \mathcal{E})$, the loss of the initial set, and it decreases only finitely often, after which k kicks end the algorithm. Finally, Cor. 3.19 turns the uncritical set Ψ^* into a BCNF design of (R, Σ) with enforced FDs Ψ^{*+}, whose loss is $w(\Sigma_a - \Psi^{*+})$ by definition, at any time. □

Proof of Theorem A.11. No source A is the right-hand side of a non-trivial FD of Σ^+, since such an FD would yield a minimal FD of Σ with right-hand side A. Let $\mathcal{E} \subseteq \Sigma^+$ be closed, and suppose that every minimal FD $X \to A$ of $\mathcal{E}$ satisfies $XA \subseteq R_P$ for the part $P \ni A$. We first show that $Y^+_{\mathcal{E}} \cap P_i = (Y \cap R_i)^+_{\mathcal{E}} \cap P_i$ for every $Y \subseteq R$. The closure of Y under the minimal FDs of $\mathcal{E}$ adds an attribute of P_i only through a minimal FD whose left-hand side lies in R_i, whose attributes are sources, which belong to Y, or attributes of P_i, which by induction lie in $(Y \cap R_i)^+_{\mathcal{E}}$. Consequently, $\mathcal{E}_i = \mathcal{E}[R_i]$ satisfies $X^+_{\mathcal{E}_i} = X^+_{\mathcal{E}} \cap R_i$ for $X \subseteq R_i$, the minimal FDs of $\mathcal{E}$ with right-hand side in P_i are the minimal FDs of $\mathcal{E}_i$ with right-hand side in P_i, and for each of them, $X \to A$, we have $\mathcal{E}[XA] = \mathcal{E}_i[XA]$, so it is uncritical for $\mathcal{E}$ iff it is uncritical for $\mathcal{E}_i$. The same argument for Σ, whose minimal FDs satisfy the assumption by the definition of the parts, shows that $\Sigma^+_i = \Sigma^+[R_i]$ and that the minimal FDs of Σ with right-hand side in P_i are the minimal FDs of (Σ_i).

For the upper bound, let $\mathcal{E}_i$ be enforceable for (R_i, Σ_i) with loss $\lambda_w(R_i, \Sigma_i)$, and let U_i be its uncritical minimal FDs, which form an uncritical set that generates $\mathcal{E}_i$ by Thm. 3.18. Let $\mathcal{E} = (U_1 \cup \cdots \cup U_r)^+$. Every minimal FD of $\mathcal{E}$ with right-hand side in P_i has its left-hand side in R_i by the argument above applied to the generators, so $\mathcal{E}$ satisfies the assumption, $\mathcal{E}[R_i] = \mathcal{E}_i$, and every element of U_i is an

uncritical minimal FD of $\mathcal{E}$. Hence, $\mathcal{E}$ is enforceable by Cor. 3.19, it is the set of enforced FDs of a part-respecting BCNF design, and a minimal FD of Σ with right-hand side in P_i belongs to $\mathcal{E}$ iff it belongs to $\mathcal{E}_i$, so the loss of the design is $\sum_i \lambda_w(R_i, \Sigma_i)$. If every $\mathcal{E}_i$ is plain, then $U_1 \cup \cdots \cup U_r$ may be replaced by the minimal FDs of Σ in $\mathcal{E}_1 \cup \cdots \cup \mathcal{E}_r$, and the design is plain.

For the lower bound, let $\mathcal{E}$ be the enforced FDs of a part-respecting BCNF design. Every FD $X \to A$ of $\mathcal{E}_i = \mathcal{E}[R_i]$ with $A \in P_i$ is implied by the uncritical minimal FDs of $\mathcal{E}$, and the closure of X under them adds attributes of P_i only through uncritical minimal FDs of $\mathcal{E}$ with right-hand side in P_i, which are uncritical minimal FDs of $\mathcal{E}_i$. Hence, $\mathcal{E}_i$ is enforceable for (R_i, Σ_i) by Thm. 3.18, and the loss of the design is the sum of the losses $w((\Sigma_i)_a - \mathcal{E}_i) \geq \lambda_w(R_i, \Sigma_i)$. If the design is plain, so is $\mathcal{E}_i$, since $\mathcal{E}_i$ is generated by the minimal FDs of Σ in $\mathcal{E}$ with right-hand side in P_i. Together with the upper bound, this proves both equalities. □

Proof of Corollary A.12. The sources and the parts are computed from Σ_a in polynomial time, and $(\Sigma_i)_a$ is the set of elements of Σ_a with right-hand side in P_i. Alg. 1, restricted to $W = \emptyset$ for plain designs, solves (R_i, Σ_i) in time bounded by a function of $|R_i|$, since $|(\Sigma_i)_a| \leq |R_i| \cdot 2^{|R_i|}$, and the glued design of the proof of Thm. A.11 is computed in polynomial time from the designs of the parts. □

Proof of Proposition A.6. Let $S_1, \ldots, S_m$ be the critical schemata of $\mathcal{D}_0$ and let $T = R - (S_1 \cup \cdots \cup S_m)$, so that $S_1, \ldots, S_m, T$ partition R and every schema of $\mathcal{D}_0$ other than a key schema lies inside one part. The projection of Σ onto a minimal key K of (R, Σ) is trivial, since an FD $Y \to B$ with $Y \subset K$ and $B \in K - Y$ would make $K - B$ a key. Since $\mathcal{D}_0$ is dependency-preserving for Σ, the set Σ^+ is therefore the closure of $\Sigma[S_1] \cup \cdots \cup \Sigma[S_m] \cup \Sigma[T]$, and the attribute closure of a set inside a part stays inside the part. By Lem. 4.2, applied m times, $\lambda_w(R, \Sigma) = \sum_{i=1}^{m} \lambda_w(S_i, \Sigma[S_i]) + \lambda_w(T, \Sigma[T])$, where the weighted version of the lemma holds with the same proof. The schemata of $\mathcal{D}_0$ inside T are in BCNF and their projections generate $\Sigma[T]$, and $K \cap T$ is a minimal key of $(T, \Sigma[T])$ since closures stay inside the parts, so these schemata and $K \cap T$ form a lossless, dependency-preserving decomposition of $(T, \Sigma[T])$ into BCNF, $\lambda_w(T, \Sigma[T]) = 0$ by Thm. 3.15, and the minimal keys of the schemata inside T generate $\Sigma[T]$. Alg. 3 adds to Γ these key dependencies, the trivial key dependency of a key schema if $\mathcal{D}_0$ has one, and, for every S_i, the key dependencies of a design of $(S_i, \Sigma[S_i])$ with loss $\lambda_w(S_i, \Sigma[S_i])$ by Thm. 5.5, whose enforced FDs $\mathcal{E}_i$ are enforceable for $(S_i, \Sigma[S_i])$. As in the proof of Lem. 4.2, the closure Γ^+ of the union of the enforceable sets $\mathcal{E}_1, \ldots, \mathcal{E}_m, \Sigma[T]$ over attribute-disjoint schemata is enforceable for (R, Σ), so line 9 succeeds by Thm. 3.18, and $w(\Sigma_a - \Gamma^+) = \sum_i w((\Sigma[S_i])_a - \mathcal{E}_i) = \lambda_w(R, \Sigma)$. The design of line 11 has loss at most $w(\Sigma_a - \Gamma^+)$ by Prop. 5.8 and at least $\lambda_w(R, \Sigma)$ by definition, which proves the claim. □

Proof of Theorem 5.9. Every element of Γ is a key dependency of a key design of some $(S, \Sigma[S])$, so $\Gamma \subseteq \Sigma^+$. If line 9 returns a lossless, dependency-preserving decomposition $\mathcal{D}$ of (R, Γ^+) into BCNF, then $(\mathcal{D}, \kappa)$ is a key design of (R, Σ) since every minimal key of $(S, \Gamma^+[S])$ is a key of S for Σ, its enforced FDs are Γ^+ since $\mathcal{D}$ is dependency-preserving and in BCNF for Γ^+, and it is sound and locally enforced as in the proof of Thm. 5.5. Its loss is $w(\Sigma_a - \Gamma^+)$ by definition, and the uncritical design is a BCNF design of (R, Σ) with loss $w(O)$ by Thm. 3.22, so line 11 returns a BCNF design whose loss is at most the minimum of the two by Prop. 5.8. Otherwise, the claim follows from Thm. 5.5. □